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Normal Distribution

Moments About Origin



1st Moment

$$\text{P.d.f} = p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\begin{aligned} E(x) &= x \cdot p(x) \\ &= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \end{aligned}$$

$$\begin{aligned} \text{Let } \frac{x-\mu}{\sigma} &= z, \quad dx = \sigma dz \\ \text{and } x &= \mu + \sigma z \end{aligned}$$

By putting values we get

$$\begin{aligned} &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z) \cdot e^{-\frac{z^2}{2}} \cdot \sigma dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mu e^{-\frac{z^2}{2}} + \sigma z \cdot e^{-\frac{z^2}{2}} dz \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} u e^{-z^2/2} dz + \underbrace{\int_{-\infty}^{\infty} f(z) e^{-z^2/2} dz}_{\text{odd}} \right)$$

$$= \frac{2u}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz$$

$$\text{Let } z^2/2 = p \quad , \quad 2z dz = dp$$

$$z = \sqrt{2p} \quad , \quad dz = dp/z$$

$$dz = dp/\sqrt{2p}$$

By putting values we get.

$$= \frac{2u}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-p} dp/\sqrt{2p}$$

$$= \frac{2u}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-p} dp \cdot (\sqrt{2p})^{-1}$$

$$= \frac{2u}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p^{-1/2} \cdot e^{-p} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{2u}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p^{-1/2} \cdot e^{-p}$$

$$E(X) = \frac{\mu}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$$= \frac{\mu}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$$E(X) = \mu$$

2nd Moment

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Let: } \frac{x-\mu}{\sigma} = z, \quad dx = \sigma dz$$

$$\text{and } x = \mu + \sigma z$$

By putting these values we get

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z)^2 e^{-z^2/2} \cdot \sigma dz$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mu^2 \cdot e^{-z^2/2} dz + \int_{-\infty}^{\infty} \sigma^2 z^2 \cdot e^{-z^2/2} dz + 2 \underbrace{\int_{-\infty}^{\infty} \sigma z e^{-z^2/2} dz}_{\text{odd}}$$

$$= \frac{2\mu^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz + \frac{2\delta^2}{\sqrt{2\pi}} \int_0^{\infty} e^{-z^2/2} \cdot z^2 dz$$

Let : $z^2/2 = p$, $z dz = dp$
 $z = \sqrt{2p}$, $dz = \frac{dp}{z} = \frac{dp}{\sqrt{2p}}$

By putting values we get.

$$= \frac{2\mu^2}{\sqrt{2\pi}} \int e^p \frac{dp}{\sqrt{2p}} + \frac{2\delta^2}{\sqrt{2\pi}} \int z^2 \cdot e^p \frac{dp}{\sqrt{2p}}$$

$$= \frac{2\mu^2}{\sqrt{2\pi}} \int e^p \frac{dp}{\sqrt{2p}} + \frac{2\delta^2}{\sqrt{2\pi}} \int p^{-1/2} e^p \cdot 2p \cdot dp$$

$$= \frac{2\mu^2}{\sqrt{2\pi}} \int p^{-1/2} \cdot e^p \cdot dp \cdot \frac{1}{\sqrt{2}} + \frac{2\delta^2}{\sqrt{2\pi}} \int 2p^{1/2} \cdot e^p \cdot dp$$

$$= \frac{2\mu^2}{2\sqrt{\pi}} \int p^{-1/2} \cdot e^p \cdot dp + \frac{2\delta^2}{2\sqrt{\pi}} \cdot 2 \int p^{1/2} \cdot e^p \cdot dp$$

$$= \frac{\mu^2}{\sqrt{\pi}} \cdot \sqrt{\pi} + \frac{2\delta^2}{\sqrt{\pi}} \cdot \frac{1}{2} \sqrt{\pi}$$

$$E(x^2) = \mu^2 + \delta^2$$

3rd Moment

$$E(X^3) = \int_{-\infty}^{\infty} x^3 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^3 \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

let, $z = \frac{x-\mu}{\sigma}$, $dx = \sigma dz$

and $x = \mu + \sigma z$

By putting the values we get.

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z)^3 e^{-z^2/2} \sigma dz$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} \mu^3 e^{-z^2/2} dz + 3 \underbrace{\int_{-\infty}^{\infty} \mu^2 \sigma z e^{-z^2/2} dz}_{\text{odd}} \right]$$

$$+ 3 \int_{-\infty}^{\infty} \mu \sigma^2 z^2 e^{-z^2/2} dz + \underbrace{\int_{-\infty}^{\infty} \sigma^3 z^3 e^{-z^2/2} dz}_{\text{odd}}$$

$$= \frac{2\mu^3}{\sqrt{2\pi}} \int_0^{\infty} z^{-3/2} dz + \frac{6\delta^2\mu}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-z^2/2} dz$$

Now let: $z^2/2 = p$, $z = \sqrt{2p}$

and $z dz = dp = z dz$

$$dz = \frac{dp}{z} = \frac{dp}{\sqrt{2p}}$$

By putting the values we get.

$$= \frac{2\mu^3}{\sqrt{2\pi}} \int_0^{\infty} e^{-p} \frac{dp}{\sqrt{2p}} + \frac{6\delta^2\mu}{\sqrt{2\pi}} \int_0^{\infty} 2p e^{-p} \frac{dp}{\sqrt{2p}}$$

$$= \frac{2\mu^3}{\sqrt{2\pi}} \int_0^{\infty} p^{-1/2} e^{-p} \cdot dp \cdot \frac{1}{\sqrt{2}} + \frac{2 \times 6\delta^2\mu}{\sqrt{2\pi}} \int_0^{\infty} p^{1/2} e^{-p} \cdot dp$$

$$= \frac{2\mu^3}{2\sqrt{\pi}} \int_0^{\infty} p^{-1/2} e^{-p} \cdot dp + \frac{2 \times 6\delta^2\mu}{2\sqrt{\pi}} \int_0^{\infty} p^{1/2} e^{-p} \cdot dp$$

$$= \frac{\mu^3}{\sqrt{\pi}} \int_0^{\infty} p^{-1/2} e^{-p} dp + \frac{6\delta^2\mu}{\sqrt{\pi}} \int_0^{\infty} p^{1/2} e^{-p} \cdot dp$$

$$= \frac{\mu^3}{\sqrt{\pi}} \cdot \sqrt{\pi} + \frac{6\delta^2\mu}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \sqrt{\pi}$$

$$E(x^3) = \mu^3 + 3\delta^2\mu$$

4th Moment

$$E(X^4) = \int_{-\infty}^{\infty} x^4 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^4 \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

let: $z = \frac{x-\mu}{\sigma}$, $dx = \sigma dz$

and $x = \mu + \sigma z$

By putting the values we get.

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z)^4 \cdot e^{-z^2/2} \cdot \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} \mu^4 e^{-z^2/2} dz + 4 \int_{-\infty}^{\infty} \mu^3 \sigma z e^{-z^2/2} dz + 6 \right.$$

$$\left. \int_{-\infty}^{\infty} \mu^2 \sigma^2 z^2 e^{-z^2/2} dz + 4 \int_{-\infty}^{\infty} \mu \sigma^3 z^3 e^{-z^2/2} dz + \right.$$

$$\left. \int_{-\infty}^{\infty} \sigma^4 z^4 e^{-z^2/2} dz \right] \frac{1}{\sqrt{2\pi}}$$

$$= \frac{2u^4}{\sqrt{2\pi}} \int_0^{\infty} e^{-z^2/2} dz + \frac{12u^2\delta^2}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-z^2/2} dz$$

$$+ \frac{2\delta^4}{2\pi} \int_0^{\infty} z^4 e^{-z^2/2} dz$$

let: $z^2 = 2p \rightarrow z = \sqrt{2p}, dz = dp/\sqrt{2p}$

By putting these values we get:-

$$= \frac{2u^4}{\sqrt{2\pi}} \int_0^{\infty} e^{-p} \frac{dp}{\sqrt{2p}} + \frac{12u^2\delta^2}{\sqrt{2\pi}} \int_0^{\infty} 2p e^{-p} \frac{dp}{\sqrt{2p}}$$

$$+ \frac{2\delta^4}{\sqrt{2\pi}} \int_0^{\infty} 4p^2 e^{-p} \frac{dp}{\sqrt{2p}}$$

$$= \frac{2u^4}{2\sqrt{\pi}} \int_0^{\infty} e^{-p} p^{-1/2} dp + \frac{2 \times 12u^2\delta^2}{2\sqrt{\pi}} \int_0^{\infty} p^{1/2} e^{-p} dp$$

$$+ \frac{2\delta^4 4}{2\sqrt{\pi}} \int_0^{\infty} p^{3/2} e^{-p} dp$$

$$= \frac{u^4}{\sqrt{\pi}} \cdot \sqrt{\pi} + \frac{6u^2\delta^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \sqrt{\pi}$$

$$+ \frac{4\delta^4}{\sqrt{\pi}} \cdot \frac{3}{4} \cdot \sqrt{\pi}$$

$$E(Y^4) = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

Moment About Mean

1st $\Rightarrow$ Moment:-

$$\begin{aligned}\mu_1 &= \mu'_1 - \mu'_1 = 0 \\ &= \mu - \mu \\ &= \mu - \mu \\ &= 0\end{aligned}$$

2nd Moment:-

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= \mu^2 + \sigma^2 - \mu^2 \\ &= \mu^2 - \mu^2 + \sigma^2 \\ &= \sigma^2\end{aligned}$$

3rd Moment:-

$$\begin{aligned}\mu_3 &= \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3 \\&= \mu^3 + 3\sigma^2\mu - 3(\mu^2 + \sigma^2)(\mu) + 2\mu^3 \\&= \mu^3 + 3\sigma^2\mu - 3\mu^3 - 3\mu\sigma^2 + 2\mu^3 \\&= 3\mu^3 - 3\mu^3 + 3\mu\sigma^2 - 3\mu\sigma^2 = 0 \\&= 0\end{aligned}$$

4th Moment:-

$$\begin{aligned}\mu_4 &= \mu_4' - 4\mu_1'\mu_3' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4 \\&= \mu^4 + 6\sigma^2\mu^2 + 3\sigma^4 - 4(\mu^3 + 3\sigma^2\mu)(\mu) + 6(\mu^2 + \sigma^2)(\mu^2) - 3\mu^4 \\&= \mu^4 + 6\sigma^2\mu^2 + 3\sigma^4 - 4\mu^4 - 12\sigma^2\mu^2 + 6\mu^4 + 6\mu^2\sigma^2 - 3\mu^4 \\&= \mu^4 - 4\mu^4 + 6\mu^4 - 3\mu^4 + 6\sigma^2\mu^2 - 12\sigma^2\mu^2 + 6\mu^2\sigma^2 + 3\sigma^4 \\&= 3\sigma^4\end{aligned}$$

$$\mu_4 = 3\sigma^4$$

Skewness

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

$$= \frac{0}{(8^2)^{3/2}}$$

$$\text{Skewness} = 0$$

Kurtosis

$$\text{Kurtosis} = \frac{\mu_4}{\mu_2^2}$$

$$= \frac{38^4}{(8^2)^2}$$

$$= \frac{38^4}{8^4}$$

$$\text{Kurtosis} = 3$$

Mean Moment Generating Function Normal Distribution

$$E(e^{t(x-\mu)}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot e^{t(x-\mu)} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} e^{tx} \cdot e^{-\mu t} dx$$

$$= \frac{1}{\sqrt{2\pi\sigma}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} e^{tx} \cdot e^{-\mu t} dx$$

$$= e^{\mu t - \frac{1}{2}\sigma^2 t^2} \cdot e^{-\mu t}$$

$$MM_0(t) = e^{\mu t - \frac{1}{2}\sigma^2 t^2} \cdot e^{-\mu t}$$

$$MM'_0(t) = e^{\frac{1}{2}\sigma^2 t^2} \cdot (\sigma^2 t)$$

$$\text{put } t=0$$

$$= e^{\frac{1}{2}\sigma^2 \cdot 0} \cdot 0$$

$$= 0$$

2nd sol:

$$\begin{aligned} MM_0''(t) &= e^{\frac{1}{2}\delta^2 t} \cdot \delta^2 t + e^{\frac{1}{2}\delta^2 t} \cdot \delta^2 \\ &= \delta^2 e^{\frac{1}{2}\delta^2 t} (t+1) \end{aligned}$$

$$\begin{aligned} \text{put } t=0 \\ &= \delta^2 (1) (0+1) \\ &= \delta^2 (1) \\ &= \delta^2 \end{aligned}$$

3rd sol:-

$$\begin{aligned} MM_0''(t) &= \left(e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 t + e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 \right) + \left(e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 + e^{\frac{1}{2}\delta^2 t^2} \right) (0) \\ &= e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 [t+1] + e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 \\ &= e^{\frac{1}{2}\delta^2 t^2} \delta^2 [t+1+1] \\ &= e^{\frac{1}{2}\delta^2 t^2} \delta^2 [t+2] \end{aligned}$$

$$\begin{aligned} \text{put } t=0 \\ &= (1) \delta^2 (0+2) \\ &= 2\delta^2 \end{aligned}$$

4th:

$$e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 t + 2e^{\frac{1}{2}\delta^2 t^2} \delta^2$$

$$\begin{aligned} NM_0(t) &= \left[e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 t + e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 \right] + 2 \left[e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 + 0 \right] \\ &= e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 [t+1] + 2e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 \\ &= e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 (t+1+2) \\ &= e^{\frac{1}{2}\delta^2 t^2} \cdot \delta^2 (t+3) \end{aligned}$$

put $t=0$

$$\begin{aligned} &= (1) \cdot \delta^2 (0+3) \\ &= 3\delta^2 \end{aligned}$$

Median of Normal Distribution

Median:

$$F(x) = \Pr(X \leq x)$$

$$= \int_{-\infty}^x f(x) \cdot dx$$

$$\text{Median} = \int_{-\infty}^m f(x) \cdot dx = 1/2$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^m e^{-1/2 \left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{let } \frac{x-\mu}{\sigma} = z, \quad dx = \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\frac{m-\mu}{\sigma}} e^{-z^2/2} \sigma dz = 1/2$$

$$\text{if } x = -\infty, \quad z = -\infty$$

$$\text{if } x = m, \quad z = \frac{m-\mu}{\sigma}$$

$$\text{So, for median, } \frac{m-\mu}{\sigma} = 0$$

$$m - \mu = 0$$

$$\boxed{m = \mu}$$

Mode of Normal Distribution.

Mode

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\ln f = \ln c - \frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2$$

Derivative w.r.t. (x) we get

$$= 0 - \frac{x}{\sigma} \left(\frac{x-\mu}{\sigma}\right) \quad (1)$$

$$\frac{x}{\sigma} \left(\frac{x-\mu}{\sigma}\right) = 0$$

$$-x + \mu = 0$$

$$\boxed{x = \mu}$$

Poisson Distribution

Moments About Origin

1st Moment:-

As we know that:

$$\mu_1 = E(X) = \sum_{x=0}^{\infty} x \cdot p(x)$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{e^{-\mu} \mu^x}{x!}$$

$$= e^{-\mu} \sum_{x=0}^{\infty} \frac{x \mu^x}{x!}$$

$$= e^{-\mu} \sum_{x=1}^{\infty} \frac{\mu^x}{(x-1)!}$$

By multiplying and dividing

μ we get

$$= e^{-\mu} \sum_{x=1}^{\infty} \frac{\mu^x}{(x-1)!} \cdot \frac{\mu}{\mu}$$

$$= e^{-\mu} \cdot \mu \sum_{x=1}^{\infty} \frac{\mu^{x-1}}{(x-1)!} \cdot \mu^{x-1}$$

$$\mu'_1 = e^{-\mu} \cdot \mu \sum_{x=0}^{\infty} \frac{\mu^{x-1}}{(x-1)!}$$

Let: $y = x - 1$

Now:

$$= e^{-\mu} \cdot \mu \sum_{y=0}^{\infty} \frac{\mu^y}{(y)!}$$

$$= e^{-\mu} \cdot \mu \cdot e^{\mu}$$

$$= e^{-\mu+\mu} \cdot \mu$$

$$= \mu \cdot e^0$$

$$\mu'_1 = \mu \quad \text{As } e^0 = 1$$

$$\mu'_1 = \mu$$

2nd Moment:

As we know that.

$$\mu'_2 = E(X^2) = \sum_{x=0}^{\infty} x^2 \cdot p(x)$$

$$\text{Here: } x^2 = x^2 - x + x$$

$$= x(x-1) + x$$

By applying expectation we get.

$$\begin{aligned}
 & E(x(x-1) + x) \\
 &= E(x(x-1)) + E(x) \\
 &= E(x(x-1)) + \lambda \quad \text{--- (1)}
 \end{aligned}$$

Now,

$$E(x(x-1)) = \sum_{x=0}^{\infty} x(x-1) p(x)$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x(x-1)(x-2)!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x(x-1)(x-2)!}$$

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!}$$

Multiplying and divided by λ^2

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} \cdot \frac{\lambda^2}{\lambda^2}$$

$$= e^{-\lambda} \cdot \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} \cdot e^{\lambda}$$

$$\sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!}$$

$$= e^{\lambda}$$

Let $x-2 = y$ So, we get

$$E(x(x-1)) = e^{-\mu} \cdot \mu^2 \sum_{y=0}^{\infty} \frac{\mu^y}{(y)!} \cdot e^{\mu}$$

$$= e^{-\mu} \cdot e^{\mu} \cdot \mu^2 \sum_{y=0}^{\infty} \frac{\mu^y}{(y)!}$$

$$= e^{-\mu+\mu} \cdot \mu^2$$

$$= \mu^2 \cdot e^0$$

$$E(x(x-1)) = \mu^2 \quad \text{As } e^0 = 1$$

$$E(x(x-1)) = \mu^2$$

$$\mu'_2 = \mu^2 + \mu \quad \text{from eqn. (A)}$$

3rd Moment:-

As we know that.

$$\mu'_3 = E(x^3) = \sum_{x=0}^{\infty} x^3 \cdot p(x)$$

$$\text{Here : } x^3 = x(x-1)(x-2) + 3x(x-1) + x$$

By applying expectations

$$\mu'_3 = E(x(x-1)(x-2) + 3x(x-1) + x)$$

$$= E(x(x-1)(x-2)) + 3E(x(x-1)) + E(x)$$

$$u_1 = E(x(x-1)(x-2)) + 3u_0^2 + u_1 - (n)$$

Now we solve $E(x(x-1)(x-2))$

So,

$$E(x(x-1)(x-2)) = \sum_{n=0}^{\infty} x(x-1)(x-2) \frac{e^{-u} u^n}{n!}$$

$$= e^{-u} \sum_{n=0}^{\infty} x(x-1)(x-2) \frac{u^n}{n!}$$

$$= e^{-u} \sum_{n=0}^{\infty} u(u+1)(u+2) \frac{u^n}{(n+3)!}$$

$$= e^{-u} \sum_{n=0}^{\infty} \frac{u^n}{(n+3)!}$$

Multiply and divided by u^3

$$= e^{-u} \sum_{n=0}^{\infty} \frac{u^n}{(n+3)!} \cdot \frac{u^3}{u^3}$$

$$= e^{-u} \cdot u^3 \sum_{n=0}^{\infty} \frac{u^n \cdot u^{-3}}{(n+3)!}$$

$$= e^{-u} \cdot u^3 \sum_{n=0}^{\infty} \frac{u^{n-3}}{(n+3)!}$$

Let $n+3 = y$ So, we get

$$= e^{-u} u^3 \sum_{y=0}^{\infty} \frac{u^y}{y!}$$

$$= e^{-u} \cdot u^3 \cdot e^u$$

$$= e^{-u+u} \cdot u^3$$

$$= e^0 \cdot u^3$$

$$E(x(x-1)(x-2)) = u^3 \quad \text{As } e^0 = 1$$

Now by putting value in equ. (A)

$$u'_3 = u^3 + 3u^2 + u$$

$$u'_3 = u^3 + 3u^2 + u$$

4th Moment:

As we know that

$$u'_4 = E(x^4) = \sum_{x=0}^{\infty} x^4 \cdot p(x)$$

$$u'_4 = x(x-1)(x-2)(x-3) + 6x(x-1)(x-2)$$

$$+ 7(x(x-1)) + x$$

By applying expectations

$$H'_4 = E(x(x-1)(x-2)(x-3)) + 6E(x(x-1)(x-2)) + 7E(x(x-1)) + E(x). \quad \text{--- (A)}$$

$$H'_4 = E(x(x-1)(x-2)(x-3)) + 6H^3 + 7H^2 + H \quad \text{--- (2)}$$

Now, we solve $E(x(x-1)(x-2)(x-3))$

$$\begin{aligned} E(x(x-1)(x-2)(x-3)) &= \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \frac{e^{-x} x^n}{n!} \\ &= e^{-x} \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \frac{x^n}{n!} \\ &= e^{-x} \sum_{x=4}^{\infty} \frac{x^n}{(x-4)!} \end{aligned}$$

Multiply and divided by x^4

$$= e^{-x} \sum_{x=4}^{\infty} \frac{x^n}{(x-4)!} \cdot \frac{x^4}{x^4}$$

$$= e^{-x} \cdot x^4 \sum_{x=4}^{\infty} \frac{x^{n-4}}{(x-4)!}$$

Let $y = x-4$ So, we get.

$$E(x(x-1)(x-2)(x-3)) = e^{-\mu} \cdot \mu^4 \sum_{j=0}^{\infty} \frac{\mu^j}{j!}$$

$$= e^{-\mu} \cdot \mu^4 \cdot e^{\mu}$$

$$= e^{-\mu + \mu} \cdot \mu^4$$

$$= e^0 \cdot \mu^4$$

$$E(x(x-1)(x-2)(x-3)) = \mu^4$$

By putting in equ. (B) we get

$$\mu'_4 = \mu'' + 6\mu^3 + 7\mu^2 + \mu$$

Moments About Mean

1st Moment

$$\mu_1 = \mu'_1$$

$$\text{As } \mu'_1 = \mu$$

$$\text{So, } \mu_1 = \mu$$

$$\mu_1 = \mu$$

2nd Moment:-

As we know that:

$$\mu_2 = \mu_2' - (\mu_1')^2$$

As $\mu_1' = \mu$
So, $\mu_2 = \mu_2' - (\mu)^2$ and $\mu_1' = \mu + \mu^2$
 $= \mu + \mu^2 - \mu^2$

$$= \mu + \mu^2 - \mu^2$$

$$\mu_2 = \mu$$

$$\mu_2 = \mu$$

3rd Moment:-

As we know that:

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$$

$$\text{As } \mu_2' = \mu^2 + 3\mu\mu_1'$$

$$\mu_1' = \mu + \mu^2$$

$$\mu_1' = \mu$$

So,

$$\mu_3 = \mu^3 + 3\mu^2 + \mu - 3(\mu)(\mu + \mu^2) + 2\mu^3$$

$$\begin{aligned}
 &= \mu^3 + 3\mu^2 + \mu - 3\mu^3 - 2\mu^2 - 2\mu \\
 &= \mu^3 - 3\mu^3 + 2\mu^2 + 2\mu - 3\mu - \mu \\
 &= -2\mu^3 + 2\mu^2 + \mu
 \end{aligned}$$

$$\mu_3 = \mu$$

$$\mu_3 = \mu$$

4th Moment:-

As we know that:-

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6(\mu_2')^2(\mu_1') - 3(\mu_1')^4$$

$$\text{As: } \mu_4' = \mu^4 + 6\mu^3 + 7\mu^2 + \mu$$

$$\mu_3' = \mu^3 + 3\mu^2 + \mu$$

$$\mu_2' = \mu^2 + \mu$$

$$\mu_1' = \mu$$

By putting values we get.

$$\mu_4 = (\mu^4 + 6\mu^3 + 7\mu^2 + \mu) - 4\mu^3(\mu^2 + \mu) + 6\mu^2(\mu^3 + 3\mu^2 + \mu) - 3\mu^4$$

$$u_4 = u'''' + 6u''' + 7u'' + u - 4u' - 12u^3 - 4u^2$$

$$+ 16u^4 + 6u^5 - 3u^6$$

$$= u'''' - 4u'' + 6u' - 3u' + 6u^3 - 12u^3 + 6u^2$$

$$+ 7u^2 - 4u^2 + u$$

$$= -7u'' + 7u'' + 12u^3 - 12u^3 + 3u^2 + u$$

$$u_4 = 3u^2 + u$$

$$u_4 = 3u^2 + u$$

As we see that:-

$$u_1 = u_2 = u_3 = u$$

And

$$u_4 = 3u^2 + u$$

Probability Generating Function For Poisson Distribution

$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!}$$

$$E(\theta^x) = \sum_{x=0}^{\infty} \frac{e^{-\mu} \mu^x}{x!} \theta^x$$

$$= \sum_{x=0}^{\infty} \frac{e^{-\mu} (\mu\theta)^x}{x!}$$

$$= e^{-\mu} \left[\frac{(\mu\theta)^0}{0!} + \frac{(\mu\theta)^1}{1!} + \frac{(\mu\theta)^2}{2!} + \frac{(\mu\theta)^3}{3!} + \dots \right]$$

$$= e^{-\mu} \cdot e^{\mu\theta}$$

$$= e^{-\mu(1-\theta)}$$

1st:-

$$G' = \frac{d}{d\theta} (e^{-\mu(1-\theta)})$$
$$= \mu \cdot e^{-\mu(1-\theta)} \cdot (-\mu)(-1)$$

$$\text{put } \theta = 1$$

$$= \mu e^{-\mu(1-1)}$$

$$E(X) = \mu(1) = \mu$$

2nd:-

$$G'' = \frac{d}{d\theta} (\mu \cdot e^{-\mu(1-\theta)})$$

$$= \mu \cdot e^{-\mu(1-\theta)} \cdot \mu$$

$$= \mu^2 \cdot e^{-\mu(1-\theta)}$$

$$\text{put } \theta = 1$$

$$= \mu^2 \cdot e^0$$

$$= \mu^2$$

$$E(x^2) = \mu^2$$

$$V(x) = G'' + G' - (G')^2$$

$$= \mu^2 + \mu - \mu^2$$

$$V(x) = \mu$$

3rd:

$$G''' = \frac{d}{d\theta} (\mu^2 \cdot e^{-\mu(1-\theta)})$$

$$= \mu^2 \cdot e^{-\mu(1-\theta)} (-\mu)(-1)$$

$$= \mu^2 \cdot \mu \cdot e^{-\mu(1-\theta)}$$

put $\theta = 1$

$$= \mu^3 \cdot e^{-\mu(1-1)}$$

$$= \mu^3 \cdot e^0$$

$$= \mu^3$$

4th:

$$G^{IV} = \frac{d}{d\theta} (\mu^3 \cdot e^{-\mu(1-\theta)})$$

$$= \mu^3 \cdot e^{-\mu(1-\theta)} \cdot (-\mu)(-1)$$

$$= \mu^3 \cdot \mu \cdot e^{-\mu(1-\theta)}$$

$$= \mu^4 \cdot e^{-\mu(1-1)} \quad \text{put } \theta = 1$$

$$= \mu^4 \cdot e^0$$

$$= \mu^4$$

Kurtosis

Poisson Distribution

$$\text{Kurtosis} = \frac{\mu_4}{\mu_2^2}$$

$$= \frac{3\mu^2 + \mu}{(\mu)^2}$$

$$= \frac{\mu(3\mu + 1)}{\mu^2}$$

$$= \frac{\mu(3\mu + 1)}{\mu^2}$$

$$= \frac{3\mu + 1}{\mu}$$

Skewness

Poisson Distribution

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

By putting values we get.

$$= \frac{\mu}{(\mu)^{3/2}}$$

$$= \mu / \mu \cdot \mu^{1/2}$$

$$= \mu / (\mu \cdot \mu^{1/2})$$

$$\text{Loss} = \frac{1}{\sqrt{\mu}}$$

Moments Generating Function Poisson Distribution

$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!} \quad x = 0, 1, 2, \dots$$

M.G.F of Poisson Distribution

$$\begin{aligned} M_0(t) &= E(e^{tx}) = \sum_{x=0}^{\infty} \frac{e^{-\mu} \mu^x}{x!} (e^t)^x \\ &= \sum_{x=0}^{\infty} \frac{e^{-\mu} (\mu e^t)^x}{x!} \\ &= e^{-\mu} \sum_{x=0}^{\infty} \frac{(\mu e^t)^x}{x!} \\ &= e^{-\mu} \left[1 + \mu e^t + \frac{(\mu e^t)^2}{2!} + \frac{(\mu e^t)^3}{3!} + \dots \right] \\ &= e^{-\mu} \cdot e^{\mu e^t} \\ &= e^{-\mu + \mu e^t} \\ M_0(t) &= e^{-\mu(1-e^t)} \end{aligned}$$

1st Moment:-

$$\begin{aligned} M_0'(t) &= e^{-\mu(1-e^t)} \cdot -\mu(0-e^t) \\ &= \mu e^t e^{-\mu(1-e^t)} \end{aligned}$$

$$\begin{aligned} \text{put } t=0 \\ &= \mu e^0 e^{-\mu(1-e^0)} \end{aligned}$$

$$E(x) = \mu$$

$$\because e^0 = 1$$

2nd Moment:-

$$M_0'(t) = e^{-\mu(1-e^t)}$$

$$M_0''(t) = \mu(e^t \cdot e^{-\mu(1-e^t)} - \mu(0-e^t) + e^t \cdot e^{-\mu(1-e^t)}) \\ = \mu e^t e^{-\mu(1-e^t)} [1 + \mu e^t]$$

put $t=0$

$$= \mu e^0 \cdot e^{-\mu(1-e^0)} [1 + \mu e^0] \\ = \mu(1)(1+\mu) \\ = \mu(1+\mu)$$

$$E(X^2) = \mu + \mu^2$$

3rd Moment:-

$$\mu e^t e^{-\mu(1-e^t)} [1 + \mu e^t]$$

$$M_0'''(t) = \mu(e^t \cdot e^{-\mu(1-e^t)} + e^t \cdot e^{-\mu(1-e^t)} - \mu(0-e^t)) + \mu^2[2e^t e^{-\mu(1-e^t)} + e^t \cdot e^{-\mu(1-e^t)} - \mu(0-e^t)] \\ = \mu e^t e^{-\mu(1-e^t)} [1 + \mu e^t] + \mu^2 e^t e^{-\mu(1-e^t)} [2 + \mu e^t]$$

put $t=0$

$$= \mu e^0 e^{-\mu(1-e^0)} [1 + \mu e^0] + \mu^2 e^0 e^{-\mu(1-e^0)} [2 + \mu e^0] \\ = \mu(1)[1+\mu] + \mu^2(2)[2+\mu] \quad \because e^0 = 1 \\ = \mu + \mu^2 + 2\mu^2 + \mu^3$$

$$E(X^3) = \mu + 3\mu^2 + \mu^3$$

4th Moment :-

$$\mu \left[e^t e^{-\mu(1-e^t)} + \mu e^{2t} e^{-\mu(1-e^t)} \right] + \mu^2 \left[2e^{2t} e^{-\mu(1-e^t)} + \mu e^{3t} e^{-\mu(1-e^t)} \right]$$

$$\begin{aligned} M_0^{(4)}(t) &= \left[\mu \left(e^t e^{-\mu(1-e^t)} + e^t e^{-\mu(1-e^t)} - \mu(0-e^t) \right) + \mu \left(2e^{2t} e^{-\mu(1-e^t)} - \mu(0+e^t) \right) \right] \\ &\quad + \mu^2 \left[4e^{2t} e^{-\mu(1-e^t)} + 2e^{3t} e^{-\mu(1-e^t)} - \mu(0+e^t) + \mu(3e^{3t} e^{-\mu(1-e^t)} + e^{3t} (e^{-\mu(1-e^t)} - \mu(0-e^t)) \right] \\ &= \mu \left[e^t e^{-\mu(1-e^t)} (1 + \mu e^t) + \mu e^{2t} e^{-\mu(1-e^t)} (2 + \mu e^t) \right] + \mu^2 \left[e^{2t} e^{-\mu(1-e^t)} (4 + 2\mu e^t) + \mu e^{3t} e^{-\mu(1-e^t)} (3 + \mu e^t) \right] \end{aligned}$$

put $t=0$

$$\begin{aligned} &= \mu \left[e^0 e^{-\mu(1-e^0)} (1 + \mu e^0) + \mu e^{2 \cdot 0} e^{-\mu(1-e^0)} (2 + \mu e^0) \right] + \mu^2 \left[e^{2 \cdot 0} e^{-\mu(1-e^0)} (4 + 2\mu e^0) + \mu e^{3 \cdot 0} e^{-\mu(1-e^0)} (3 + \mu e^0) \right] \\ &= \mu \left[(1)(1+\mu) + \mu(2+\mu) \right] + \mu^2 \left[(1)(4+2\mu) + \mu(3+\mu) \right] \\ &= \mu \left[(1+\mu) + 2\mu + \mu^2 \right] + \mu^2 \left[4 + 2\mu + 3\mu + \mu^2 \right] \\ &= \mu + \mu^2 + 2\mu^2 + \mu^3 + 4\mu^2 + 2\mu^3 + 3\mu^3 + \mu^4 \end{aligned}$$

$$E(X^4) = \mu + 7\mu^2 + 6\mu^3 + \mu^4$$

Moment About Mean

1st Moment :-

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$\mu_1 = \mu - \mu = 0$$

$$\mu_1 = 0$$

2nd Moment :-

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$= \mu + \mu_2' - \mu^2$$

$$= \mu + \mu_2' - \mu^2$$

$$\mu_2 = \mu$$

3rd Moment :-

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$$

$$= \mu + 3\mu^2 + \mu^3 - 3(\mu(\mu + \mu^2)) + 2\mu^3$$

$$= \mu + 3\mu^2 + \mu^3 - 3(\mu^2 + \mu^3) + 2\mu^3$$

$$= \mu + 3\mu^2 + \mu^3 - 3\mu^2 - 3\mu^3 + 2\mu^3$$

$$= \mu + 3\mu^2 - 3\mu^2 + 3\mu^3 - 3\mu^3$$

$$\mu_3 = \mu$$

4th Moment :-

$$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6(\mu_1')^2(\mu_2') - 3(\mu_1')^4$$

$$= \mu + 7\mu^2 + 6\mu^3 + \mu^4 - 4(\mu + 3\mu^2 + \mu^3)(\mu) + 6(\mu^2(\mu + \mu^2)) - 3\mu^4$$

$$= \mu + 7\mu^2 + 6\mu^3 + \mu^4 - 4(\mu^2 + 3\mu^3 + \mu^4) + 6(\mu^3 + \mu^4) - 3\mu^4$$

$$= \mu + 7\mu^2 + 6\mu^3 + \mu^4 - 4\mu^2 - 12\mu^3 - 4\mu^4 + 6\mu^3 + 6\mu^4 - 3\mu^4$$

$$= \mu + 7\mu^2 - 4\mu^2 + 6\mu^3 - 12\mu^3 + 6\mu^3 + \mu^4 - 4\mu^4 + 6\mu^4 - 3\mu^4$$

$$\mu_4 = \mu + 3\mu^2$$

Poisson Distribution

Factorial G.F. of Poisson Distribution:

$$H_0(\alpha) = e^{\mu\alpha}$$

By taking log of F.G.F we get.

$$\begin{aligned} H_0(\alpha) &= \log(e^{\mu\alpha}) \\ &= \mu\alpha \log e \end{aligned}$$

By taking Derivative w.r.t. (α) we get.

$$h'_0(\alpha) = \mu\alpha(1) + \alpha$$

$$= \mu + \alpha$$

$$\text{put } \alpha = 0$$

$$= \mu + 0$$

$$= \mu$$

$$\mu$$

$$h''_0(\alpha) = 0$$

$$h'''_0(\alpha) = 0$$

$$h^{iv}_0(\alpha) = 0$$

Comulents : (Kappa's) :

1st Cr-

$$K_1 = K_0(1) = m_1' \\ = \mu = m_1'$$

2nd Cr-

$$K_2 = K_0(1) + K_0(2) = \mu_2 \\ = \mu + 0 \\ = \mu = \mu_2$$

3rd Cr-

$$K_3 = K_0(1) + 3K_0(2) + K_0(3) = \mu_3 \\ = \mu + 3(0) + 0 \\ = \mu = \mu_3$$

4th Cr-

$$K_4 = K_0(1) + 7K_0(2) + 6K_0(3) + K_0(4) \\ = \mu + 7(0) + 6(0) + 0 \\ = \mu$$

$$\mu_4 = K_4 + 3K_2^2 \\ = \mu + 3(\mu)^2 \\ = \mu + 3\mu^2$$

Moment Generating Function of

Poisson Distribution:

$$M_0(t) = e^{-\mu(1-e^t)}$$

By taking log of M.G.F of Poisson Distribution we get.

$$\log M_0(t) = \log (e^{-\mu(1-e^t)})$$

$$C_0(t) = -\mu(1-e^t)$$

By taking derivative w.r.t (t) we get.

1st:

$$C'(t) = -u(1 - e^t)(1)$$

$$\text{put } t = 0$$

$$= -u(1 - 1)$$

$$= -u(0)$$

$$= 0$$

2nd:

$$C''(t) = -u(-e^t)$$

$$C''_0(t) = -u(-e^t)$$

$$= ue^t$$

$$\text{put } t = 0$$

$$= u$$

3rd:

$$ue^t$$

$$C'''(t) = ue^t$$

$$\text{put } t = 0$$

$$= u$$

4th:

$$ue^t$$

$$C^{(iv)}(t) = ue^t$$

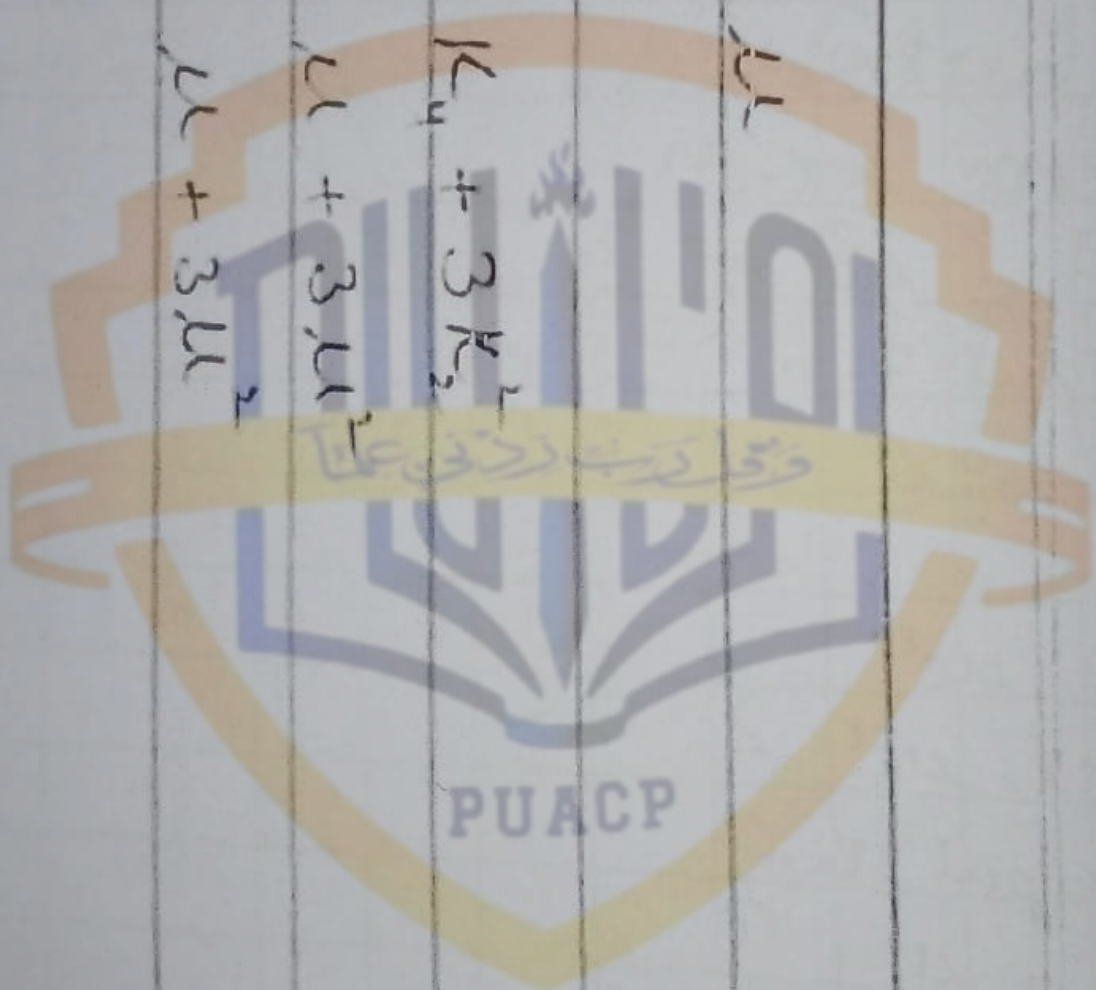
$$\text{put } t = 0$$

$$= \mu$$

$$m_u = \mu_1 + 3\mu_2^2$$

$$= \mu + 3\mu^2$$

$$m_u = \mu + 3\mu^2$$



Factorial Generating Function Poisson Distribution

$$M_0(t) = e^{-\mu + \mu e^t}$$

$$\text{put } e^t = 1 + \alpha$$
$$= e^{-\mu + \mu(1 + \alpha)}$$

$$= e^{-\mu + \mu + \mu\alpha}$$

$$M_0(t) = e^{\mu\alpha}$$

1st: $M_0'(t) = e^{\mu\alpha} \cdot \mu$

$$= \mu e^{\mu\alpha}$$

$$\text{put } \alpha = 0$$

$$= \mu e^0$$

$$= \mu$$

$$\mu e^{\mu\alpha}$$

2nd: $M_0'' = \mu e^{\mu\alpha} \cdot \mu$

$$= \mu^2 e^{\mu\alpha}$$

$$\text{put } \alpha = 0$$

$$= \mu^2 e^0$$

$$= \mu^2$$

$$\mu^2 e^{i\alpha}$$

3rd $H_0'' = \mu^2 e^{i\alpha} \cdot \mu$
 $= \mu^3 e^{i\alpha}$

put $\alpha = 0$

$$= \mu^3 e^0$$

$$= \mu^3$$

$$\mu^3 e^{i\alpha}$$

4th $H_0^{iv} = \mu^3 e^{i\alpha} \cdot \mu$

$$= \mu^4 e^{i\alpha}$$

put $\alpha = 0$

$$= \mu^4 e^0$$

$$= \mu^4$$

Ordinary Moments

Moments About Origin

$$\mu_1' = E(x) = \mu$$

$$\mu_2' = E(x^2) = E[x(x-1) + E(x)]$$

$$= \mu^2 + \mu$$

$$\begin{aligned}\mu'_3 &= E(x^3) = E[x(x-1)(x-2)] + 3E[x(x-1)] + E(x) \\ &= \mu'_0 + 3\mu'_1 + \mu \\ &= \mu(\mu^2 + 3\mu + 1)\end{aligned}$$

$$\begin{aligned}\mu'_4 &= E(x^4) = E[x(x-1)(x-2)(x-3)] + 6E[x(x-1)(x-2)] + 7E[x(x-1)] + E(x) \\ &= \mu'_0 + 6\mu'_1 + 7\mu'_2 + \mu \\ &= \mu(\mu^3 + 6\mu^2 + 7\mu + 1)\end{aligned}$$

Moments About Mean.

1st:

$$\begin{aligned}\mu_1 &= \mu'_1 - \mu'_1 = 0 \\ &= \mu - \mu \\ &= 0\end{aligned}$$

2nd:

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= \mu^2 - \mu^2 \\ &= 0\end{aligned}$$

3rd:

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= \mu^3 - 3\mu^3 + 2\mu^3 \\ &= 3\mu^3 - 3\mu^3 = 0\end{aligned}$$

مثال:

$$\begin{aligned} \mathcal{L}_4 &= \mathcal{L}_4' - 4\mathcal{L}_3\mathcal{L}_1' + 6(\mathcal{L}_1')^2(\mathcal{L}_2') - 3(\mathcal{L}_1')^4 \\ &= \mathcal{L}_4' - 4\mathcal{L}_3\mathcal{L}_1' + 6\mathcal{L}_1'^2 - 3\mathcal{L}_1'^4 \\ &= 7\mathcal{L}_4' - 7\mathcal{L}_1'^4 \\ &= 0 \end{aligned}$$

Negative Binomial Distribution

Moments About Origin

1st Moment:-

$$E(x) = \sum_{x=0}^{\infty} x \cdot p(x) \quad \text{--- (A)}$$

$$\therefore p(x) = \binom{x+r-1}{x} p^r q^x \quad \text{Function}$$

$$E(x) = \sum_{x=0}^{\infty} x \binom{x+r-1}{x} p^r q^x$$

$$= \sum_{x=0}^{\infty} x \frac{(x+r-1)!}{x!(x-1)!(r-1)!} p^r q^x$$

$$\text{Let } y = x - 1$$

$$x = y + 1$$

Then:

$$E(x) = \sum_{y=0}^{\infty} \frac{(y+r)!}{y!(r-1)!} p^r q^{y+1}$$

Multiply and Dividing by r

$$E(X) = \sum_{y=0}^{r-1} r \cdot \frac{(y+r)!}{y! r(r-1)!} p^r q^y$$

$$= \sum_{y=0}^{r-1} r \binom{y+r}{y} p^r q^y$$

$$= r p^r \sum_{y=0}^{r-1} \binom{y+r}{y} q^y$$

$$= r p^r \left[\binom{r}{0} q^0 + \binom{r+1}{1} q^1 + \binom{r+2}{2} q^2 + \dots \right]$$

$$= r p^r \left[\binom{r}{0} q^0 + \binom{r+1}{1} q^1 + \binom{r+2}{2} q^2 + \dots \right]$$

$$= r p^r \left[1 + (r+1)q + \frac{(r+2)(r+1)}{2!} q^2 + \dots \right]$$

$$= r p^r [1 - q]^{-(r+1)}$$

$$= r p^r (p^{-r-1})$$

$$E(X) = \frac{r p^r}{p} = \mu'$$

2nd Moment:-

$$E(X^2) = E[X(X-1)] + E(X) \quad \text{--- (1)}$$

$$E[X(X-1)] = \sum_{x=0}^{\infty} x(x-1) p(x)$$

$$= \sum_{x=0}^{\infty} x(x-1) \binom{x+r-1}{x} p^r q^x$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{(x+r-1)!}{x!(x-2)!(r-1)!} p^r q^x$$

$$= \sum_{x=2}^{\infty} \frac{(x+r-1)!}{(x-2)!(r-1)!} p^r q^x$$

Multiply and dividing by $r(r+1)$

$$= p^r r(r+1) \sum_{x=2}^{\infty} \frac{(x+r-1)!}{(x-2)! r(r+1)(r-1)!} q^x$$

$$= p^r r(r+1) \sum_{x=2}^{\infty} \binom{x+r-1}{x-2} q^x$$

let: $y = x-2$

$$x = y+2$$

$$= p^r r(r+1) \sum_{y=0}^{\infty} \binom{y+r+1}{y} q^y q^2$$

$$= p^r q^2 r(r+1) \sum_{y=0}^{\infty} \binom{y+r+1}{y} q^y$$

$$= p^r q^2 r(r+1) \left[\binom{r+1}{0} q^0 + \binom{r+2}{1} q^1 + \binom{r+3}{2} q^2 + \dots \right]$$

$$= p^r q^2 r(r+1) [1 + (r+2)q + \dots]$$

$$= p^r q^2 r(r+1) (1-q)^{-(r+2)}$$

$$= p^r q^2 r(r+1) p^{-r-2}$$

$$= \frac{q^2 r(r+1)}{p^2}$$

By putting the values in (A) we get.

$$E(X^2) = \frac{q^2 r(r+1)}{p^2} + \frac{qr}{p} = \mu'_2$$

Variance :-

$$\text{Var} = E(x^2) - [E(x)]^2$$

$$= \frac{q^2 \cdot r(r-1)}{p^2} + \frac{qr}{p} - \frac{q^2 r^2}{p^2}$$

$$= \frac{qr}{p} \left[\frac{q(r+1) + 1}{p} - \frac{qr}{p} \right]$$

$$= \frac{qr}{p} \left[\frac{q(r+1) + p - qr}{p} \right]$$

$$= \frac{qr}{p} \left[\frac{qr + q + p - qr}{p} \right]$$

$$= \frac{qr}{p} \left[\frac{q + p}{p} \right]$$

$$= \frac{qr}{p} \left[\frac{1}{p} \right]$$

Variance : $= \frac{qr}{p^2}$

3rd Moment:

$$\mu'_3 = E(X^3)$$

$$E(X^3) = E[X(X-1)(X-2)] + 3E[X(X-1)] + E[X]$$

Now: we solve $E[X(X-1)(X-2)]$ So, we get

$$E[X(X-1)(X-2)] = \sum_{x=0}^{\infty} x(x-1)(x-2) \binom{x+r-1}{x} p^r q^x$$

$$= \sum_{x=0}^{\infty} x(x-1)(x-2) \frac{(x+r-1)!}{x!(r-1)!} p^r q^x$$

$$= \sum_{x=0}^{\infty} x(x-1)(x-2) \frac{(x+r-1)!}{x(x-1)(x-2)(x-3)!(r-1)!}$$

$$= \sum_{x=0}^{\infty} \frac{(x+r-1)!}{(x-3)!(r-1)!} p^r q^x$$

$$= \sum_{x=3}^{\infty} \frac{(x+r-1)!}{(x-3)!(r-1)!} p^r q^x$$

$$= p^r \sum_{x=3}^{\infty} \frac{(x+r-1)!}{(x-3)!(r-1)!} q^x$$

Multiply and Divided by $r(r+1)(r+2)$
 So, we get

$$= p^r \sum_{n=1}^{\infty} \frac{(n+r-1)! \dots}{(n-1)! r(r+1)(r+2)(r-1)!} q^{n+r(r+1)(r+2)}$$

$$= p^r \sum_{n=1}^{\infty} \binom{n+r-1}{n-3} q^{n+r(r+1)(r+2)}$$

$$= r(r+1)(r+2) p^r \sum_{n=1}^{\infty} \binom{n+r-1}{n-3} q^n$$

Let: $y = n-3$ and $n = y+3$

$$= r(r+1)(r+2) p^r \sum_{y=0}^{\infty} \binom{y+r+2}{y} q^{y+3}$$

$$= r(r+1)(r+2) p^r q^3 \sum_{y=0}^{\infty} \binom{y+r+2}{y} q^y$$

$$= r(r+1)(r+2) p^r q^3 \left[\binom{r+2}{0} q^0 + \binom{r+3}{1} q^1 + \binom{r+4}{2} q^2 + \dots \right]$$

$$= r(r+1)(r+2) p^r q^3 \left[1 + (r+3) q + \dots \right]$$

$$= r(r+1)(r+2) q^3 p^r (1-q)^{-(r+3)}$$

$$= r(r+1)(r+2)q^3 p^r (p)^{-(r+3)}$$

$$= r(r+1)(r+2)q^3 p^r \cdot p^r \cdot p^{-3}$$

$$E[x(x-1)(x-2)] = r(r+1)(r+2)q^3 / p^3$$

By putting this result in equ. (1)

$$E(x^3) = r(r+1)(r+2)\frac{q^3}{p^3} + 3r(r+1)\frac{q^2}{p^2} + \frac{rq}{p}$$

4th Moment:

$$E(x^4) = E[x(x-1)(x-2)(x-3)] + 6E[x(x-1)(x-2)] + 7E[x(x-1)] + E[x]$$

Now: We solve $E[x(x-1)(x-2)(x-3)]$

$$E[x(x-1)(x-2)(x-3)] = \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \cdot p(x)$$

$$= \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \binom{n+r-1}{x} p^r q^n$$

$$= \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \frac{(n+r-1)!}{x!(r-1)!} p^r q^n$$

$$= \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \frac{(n+r-1)!}{x!(r-1)!} p^r q^n$$

$$= p^r \sum_{n=4}^{\infty} \frac{(n+r-1)!}{(n-4)!(r-1)!} \cdot q^n$$

Multiply and Divided by $r(r+1)(r+2)(r+3)$

$$= p^r \sum_{n=4}^{\infty} \frac{(n+r-1)!}{(n-4)!(r-1)!(r+1)(r+2)(r+3)(r-1)!} \cdot q^n \cdot r(r+1)(r+2)(r+3)$$

$$= p^r \cdot r(r+1)(r+2)(r+3) \sum_{n=4}^{\infty} \binom{n+r-1}{n-4} \cdot q^n$$

$$= r(r+1)(r+2)(r+3) p^r \sum_{n=4}^{\infty} \binom{n+r-1}{n-4} q^n$$

Let: $y = n-4, n = y+4$

$$= r(r+1)(r+2)(r+3) p^r \sum_{y=0}^{\infty} \binom{y+r+3}{y} q^{y+4}$$

$$= r(r+1)(r+2)(r+3) p^r q^4 \left[\binom{r+3}{0} q^0 + \binom{r+4}{1} q^1 + \dots \right]$$

$$= r(r+1)(r+2)(r+3) q^4 p^r (1-q)^{-(r+4)}$$

$$= r(r+1)(r+2)(r+3) q^4 p^r p^{-(r+4)}$$

$$= r(r+1)(r+2)(r+3) q^4 p^r \cdot p^{-r} \cdot p^{-4}$$

$$= r(r+1)(r+2)(r+3)p^{-r}q^4 \therefore$$

$$E[x(x-1)(x-2)(x-3)] = r(r+1)(r+2)(r+3)q^4/p^4$$

Putting the result in (A) equation

$$E(x^4) = r(r+1)(r+2)(r+3)\frac{q^4}{p^4} + 6r(r+1)(r+2)\frac{q^3}{p^3} + 7r(r+1)\frac{q^2}{p^2} + \frac{r^2}{p}$$

Moments About Mean

1st Moment:

$$m_1 = m'_1 - m'_1 = 0$$

$$= \frac{rq}{p} - \frac{rq}{p}$$

$$= \frac{rq}{p} - \frac{rq}{p}$$

$$= 0$$

2nd Moment:

$$m_2 = m'_2 - (m'_1)^2$$

$$= r(r+1) \frac{r^2}{p^2} + \frac{r}{p} - \left(\frac{r}{p} \right)^2$$

$$= r(r+1) \frac{r^2}{p^2} + \frac{r}{p} - \frac{r^2}{p^2}$$

$$= \frac{r^2 r^2}{p^2} + \frac{r}{p} - \frac{r^2}{p^2}$$

$$= \frac{r^2 r^2}{p^2} - \frac{r^2}{p^2} + \frac{r}{p} + \frac{r}{p}$$

$$= \frac{r^2 r^2}{p^2} + \frac{r}{p}$$

$$= \frac{r}{p} \left[\frac{r}{p} + 1 \right]$$

$$= \frac{r}{p} \left[\frac{r+p}{p} \right]$$

$$= \frac{r}{p} \left[\frac{1}{p} \right]$$

$$\text{as } r+p=1$$

$$m_2 = \frac{r}{p^2}$$

3rd Moment I:

$$m_3 = m'_3 - 3m'_2m'_1 + 2(m'_1)^3$$

$$= \frac{\pi(r+1)(r+2)q^3}{p^3} + \frac{3\pi(r+1)q^2}{p^2} + \frac{\pi q}{p} - 3$$

$$\left[\left(\frac{\pi(r+1)q^2}{p^2} \right) \cdot \left(\frac{\pi q}{p} \right) \right] \left[\left(\frac{\pi q}{p} \right) \right] + \frac{2\pi^2 q^3}{p^3}$$

$$= \frac{\pi(r+1)(r+2)q^3}{p^3} + \frac{3\pi(r+1)q^2}{p^2} + \frac{\pi q}{p}$$

$$- 3 \left[\frac{\pi^2(r+1)q^3}{p^3} + \frac{3\pi^2 q^2}{p^2} + \frac{2\pi^2 q^3}{p^3} \right]$$

$$= \frac{\pi q}{p} \left[\frac{(r+1)(r+2)q^2}{p^2} + \frac{3(r+1)q}{p} + 1 \right]$$

$$= \frac{3\pi(r+1)q^2}{p^2} + \frac{3\pi q}{p} + \frac{2\pi^2 q^3}{p^3}$$

$$= \frac{\pi q}{p} \left[\frac{(r+1)(r+2)q^2 + 3(r+1)pq + p^2 \cdot 3(r+1)}{p^2} \right]$$

$$= \frac{-3prq + 2q^2r^2}{p^2}$$

$$= \frac{\pi q}{p^3} \left[\frac{(r+1)(r+2)q^2 + 3(r+1)pq + p^2 \cdot 3(r+1)}{p^2} \right]$$

$$= \frac{\pi q}{p^3} \left[\frac{q^2r^2 + 3q^2r + 3prq + 3p^2r + p^2 \cdot 3r^2 - 3prq + 2q^2r^2}{p^2} \right]$$

$$= \frac{r q}{\rho^3} \left[\rho^3 + 2 r^2 + 3 \rho r \right]$$

$$= \frac{r q}{\rho^3} \left[\rho^3 + r^2 + 2 \rho r + r^2 + \rho r \right]$$

$$= \frac{r q}{\rho^3} \left[(\rho + r)^2 + r(\rho + r) \right]$$

$$= \frac{r q}{\rho^3} \left[1 + r \right]$$

$$m_3 = \frac{r q}{\rho^3} (1 + r)$$

4th Moment:

$$m_4 = m_4' - 4 m_2' m_2' + 6 m_2' (m_2')^2 - 3 (m_2')^4$$

$$m_4 = r(r+1)(r+2)(r+3) \frac{q^4}{\rho^4} + 6 r(r+1)(r+2) \frac{q^3}{\rho^3} + 7 r(r+1) \frac{q^2}{\rho^2} + \frac{r q}{\rho}$$

$$- 4 \left[\left(r(r+1)(r+2) \frac{q^3}{\rho^3} + 3 r(r+1) \frac{q^2}{\rho^2} + \frac{r q}{\rho} \right) \left(\frac{r q}{\rho} \right) \right] + 6$$

$$\left[\left(r(r+1) \frac{q^2}{\rho^2} + \frac{r q}{\rho} \right) \left(\frac{r^2 q^2}{\rho^2} \right) \right] - 3 \left[\frac{r^4 q^4}{\rho^4} \right]$$

$$= r(r+1)(r+2)(r+3) \frac{q^4}{p^4} + 6r(r+1)(r+2) \frac{q^3}{p^3} + 7r(r+1) \frac{q^2}{p^2}$$

$$+ \frac{rq}{p} - 4 \left(\frac{r^2 q^4}{p^4} (r^2 + 3r + 2) + \frac{3r^2 q^3}{p^3} (r+1) + \frac{r^2 q^2}{p^2} \right) + 6$$

$$\left(\frac{r^3 q^4}{p^4} (r+1) + \frac{r^3 q^3}{p^3} \right) - \frac{3r^4 q^4}{p^4}$$

$$\frac{rq}{p} \cdot \left[\frac{r(r+1)(r+2)(r+3) \frac{q^3}{p^3} + 6(r+1)(r+2) \frac{q^2}{p^2} + 7(r+1) \frac{q}{p} + 1}{-4r \frac{q^3}{p^3} (r^2 + 3r + 2) - 12r \frac{q^2}{p^2} (r+1) - 4r \frac{q}{p} + 6r^2 \frac{q^3}{p^3} (r+1) + 6r^2 \frac{q^2}{p^2} - 3r^2 \frac{q}{p}} \right]$$

$$= \frac{rq}{p} \left[\frac{(r^3 + 6r^2 + 11r + 6)q^3 + 6(r^2 + 3r + 2)pq^2 + 7(r+1)p^2q + p^3 - 4rq^3}{(r^2 + 3r + 2) - 12pq^2r(r+1) - 4p^2qr + 6q^3r^2(r+1) + 6pq^2r^2 - 3q^3r^2} \right]$$

$$= \frac{rq}{p^4} \left[\frac{q^3r^3 + 6q^3r^2 + 11q^3r + 6q^3 + 6pq^2r^2 + 18pq^2r + 12pq^2 + 7p^2qr + 7p^2q + p^3}{6pq^2r^2 - 3q^3r^3 - 12q^3r^2} \right]$$

$$= \frac{rq}{p^4} \left[\frac{11q^3r + 6q^3 + 18pq^2r + 12pq^2 + 7p^2qr + 7p^2q + p^3}{-8q^3r - 12pq^2r - 4p^2qr} \right]$$

$$= \frac{rq}{p^4} \left[\frac{3q^3r + 6q^3 + 6pq^2r + 12pq^2 + 8p^2qr + 7p^2q + p^3}{-8q^3r - 12pq^2r - 4p^2qr} \right]$$

$$= \frac{r_1}{r^3} \left[r^3 + q^3 + 3r^2q + 3rq^2 + 5q^3 + 9r^2q + 6rq^2 + 3r^2p \right]$$

$$= \frac{r_1}{r^3} \left[(r+q)^3 + q(4r^2 + 9rq + 5q^2) + 3q^2(r^2 + 9rq + q^2) \right]$$

$$= \frac{r_1}{r^3} \left[1 + q \left[(2r+2q)^2 + r^2 + q^2 \right] + 3q^2 \right] \quad \text{As } r+q=1$$

$$= \frac{r_1}{r^3} \left[1 + q \left[4(r+q)^2 + q(r+q) \right] + 3q^2 \right]$$

$$= \frac{r_1}{r^3} \left[1 + q \left[4(n)^2 + q(n)^2 + 3q^2 \right] \right]$$

$$= \frac{r_1}{r^3} \left[1 + q \left[4 + q \right] + 3q^2 \right]$$

$$= \frac{r_1}{r^3} \left[1 + 4q + q^2 + 3q^2 \right]$$

So,

$$m_4 = \frac{r_1}{r^3} \left[1 + 4q + q^2 + 3q^2 \right]$$

Probability Generating Function Negative Binomial Distribution

$$P(X=x) = \binom{x+r-1}{x} p^r q^x$$

P.D.F of Negative Binomial Distribution

$$\begin{aligned} G(\theta) &= E(\theta^x) = \sum_{x=0}^{\infty} \binom{x+r-1}{x} p^r q^x \theta^x \\ &= p^r \sum_{x=0}^{\infty} \binom{x+r-1}{x} (q\theta)^x \\ &= p^r \left[1 + r q \theta + \frac{r(r+1)}{2!} (q\theta)^2 + \dots \right] \\ &= p^r (1 - q\theta)^{-r} \end{aligned}$$

$$G(\theta) = \frac{p^r}{(1 - q\theta)^r}$$

1st Moment:-

$$\begin{aligned} G'(\theta) &= p^r (-r) (1 - q\theta)^{-r-1} (0 - q) \\ &= p^r q r (1 - q\theta)^{-r-1} \\ \text{put } \theta &= 1 \\ &= p^r q r (1 - q)^{-r-1} \\ &= p^r q r p^{-r} p^{-1} \\ G'(\theta) &= \frac{qr}{p} \end{aligned}$$

2nd Moment :-

$$G''(0) = \frac{d}{d\theta} (p^2 q^2 (1 - q\theta)^{-(r+1)})$$

$$= p^2 q^2 r (1 - q\theta)^{-(r+2)} (0 - q)$$

$$= p^2 q^2 r (r+1) (1 - q\theta)^{-(r+2)}$$

$$\text{put } \theta = 1$$

$$= p^2 q^2 r (r+1) (1 - q)^{-(r+2)}$$

$$= p^2 q^2 r (r+1) p^{-r} p^{-2}$$

$$= \frac{q^2 r (r+1)}{p^2}$$

Variance :-

$$V(X) = G'' + G' - (G')^2$$

$$= \frac{q^2 r (r+1)}{p^2} + \frac{qr}{p} - \left(\frac{qr}{p}\right)^2$$

$$= \frac{q^2 r (r+1)}{p^2} + \frac{qr}{p} - \frac{q^2 r^2}{p^2}$$

$$= \frac{qr}{p} \left[\frac{q(r+1) + p}{p} - \frac{qr}{p} \right]$$

$$= \frac{qr}{p} \left[\frac{pq(r+1) + p - qr}{p} \right]$$

$$V(X) = \frac{qr}{p^2} \left[pq(r+1) + p - qr \right]$$

3rd Moment :-

$$G'''(\theta) = \frac{d}{d\theta} (p^3 q^3 r(r+1)(1-q\theta)^{r-2})$$

$$= p^3 q^3 r(r+1)(-1 \cdot 2)(1-q\theta)^{r-3} (0-q)$$

$$= p^3 q^3 r(r+1)(r+2)(1-q\theta)^{r-3}$$

$$\text{put } \theta = 1$$

$$= p^3 q^3 r(r+1)(r+2)(1-q)^{r-3}$$

$$= p^3 q^3 r(r+1)(r+2)(p^{-r-3})$$

$$= p^3 q^3 r(r+1)(r+2) p^{-r} p^{-3}$$

$$= \frac{q^3 r(r+1)(r+2)}{p^3}$$

4th Moment :-

$$G^{(4)}(\theta) = \frac{d}{d\theta} (p^3 q^3 r(r+1)(r+2)(1-q\theta)^{r-3})$$

$$= p^3 q^3 r(r+1)(r+2)(-r-3)(1-q\theta)^{r-4} (0-q)$$

$$= p^3 q^4 r(r+1)(r+2)(r+3)(1-q\theta)^{r-4}$$

$$\text{put } \theta = 1$$

$$= p^3 q^4 r(r+1)(r+2)(r+3)(1-q)^{r-4}$$

$$= p^3 q^4 r(r+1)(r+2)(r+3) p^{-r} p^{-4}$$

$$= \frac{q^4 r(r+1)(r+2)(r+3)}{p^4}$$

Moment Generating Function

Negative Binomial Distribution

$$P(X=x) = \binom{x+r-1}{x} p^r q^x$$

$$M_0(t) = E(e^{tx}) = p^r \sum_{x=0}^{\infty} \binom{x+r-1}{x} q^x e^{tx}$$

$$= p^r \sum_{x=0}^{\infty} \binom{x+r-1}{x} (qe^t)^x$$

$$= p^r \left[1 + r q e^t + \frac{r(r+1)}{2!} q^2 e^{2t} + \dots \right]$$

$$M.G.F = p^r (1 - qe^t)^{-r}$$

$$M_0(t) = \frac{p^r}{(1 - qe^t)^r}$$

1st:

$$M'_0(t) = p^r (-r) (1 - qe^t)^{-r-1} (-qe^t)$$

$$= p^r q \cdot r (1 - qe^t)^{-r-1}$$

put $t=0$

$$= p^r q r (1-q)^{-r-1}$$

$$= p^r q r (p^{-r-1})$$

$$E(X) = \frac{qr}{p}$$

2nd s-

$$e^t p^r q r (1-qe^t)^{-r-1}$$

$$p^r q r e^t (1-qe^t)^{-r-1}$$

$$M'_t = p^r q r e^t (1-qe^t)^{-r-1} + p^r q r e^t (-r-1) (1-qe^t)^{-r-2} (-qe^t)$$

$$= p^r q r e^t (1-qe^t)^{-r-1} + p^r q^2 r (r+1) e^{2t} (1-qe^t)^{-r-2}$$

put $t=0$

$$= p^r q r (1-q)^{-r-1} + p^r q^2 r (r+1) (1-q)^{-r-2}$$

$$= p^r q r (p^{-r-1}) + p^r q^2 r (r+1) (p^{-r-2})$$

$$E(X^2) = \frac{qr}{p} + \frac{q^2 r(r+1)}{p^2}$$

3rd Moment:

$$p^r q r e^t (1 - q e^t)^{-r-1} + p^r q^2 e^{2t} r(r+1) (1 - q e^t)^{-r-2}$$

$$\begin{aligned} M'''(t) &= p^r q r e^t (1 - q e^t)^{-r-1} + p^r q r e^t (-r-1) (1 - q e^t)^{-r-2} \\ &\quad (-q e^t) + p^r q^2 2 e^{2t} r(r+1) (1 - q e^t)^{-r-2} + p^r q^2 2 e^{2t} (-r-2) \\ &\quad r(r+1) (1 - q e^t)^{-r-3} (-q e^t) \\ &= p^r q r e^t (1 - q e^t)^{-r-1} + p^r q r (r+1) e^t (1 - q e^t)^{-r-2} (q e^t) \\ &\quad + 2 p^r q^2 r(r+1) e^{2t} (1 - q e^t)^{-r-2} + p^r q^2 r(r+1) (r+2) e^{2t} (1 - q e^t)^{-r-3} \\ &\quad (q e^t) \end{aligned}$$

put $t=0$

$$\begin{aligned} &= p^r q r (1 - q)^{-r-1} + p^r q^2 r(r+1) (1 - q)^{-r-2} + 2 p^r q^2 r(r+1) \\ &\quad (1 - q)^{-r-2} + p^r q^3 r(r+1) (r+2) (1 - q)^{-r-3} \\ &= p^r q r (p)^{-r-1} + p^r q^2 r(r+1) (p)^{-r-2} + 2 p^r q^2 r(r+1) \\ &\quad (p)^{-r-2} + p^r q^3 r(r+1) (r+2) (p)^{-r-3} \\ &= \frac{q}{p} r + \frac{q^2}{p^2} r(r+1) + 2 \frac{q^2}{p^2} r(r+1) + \frac{q^3}{p^3} r(r+1)(r+2) \end{aligned}$$

$$E(X^3) = \frac{q}{p} r + 3 \frac{q^2}{p^2} r(r+1) + \frac{q^3}{p^3} r(r+1)(r+2)$$

4th Moment:-

$$p^r q^r e^t (1 - qe^t)^{-r-1} + p^r q^2 r(r+1) e^{2t} (1 - qe^t)^{-r-2} \\ + 2p^r q^2 r(r+1) e^{2t} (1 - qe^t)^{-r-2} + p^r q^3 r(r+1)(r+2) \\ e^{3t} (1 - qe^t)^{-r-3}$$

$$M_0''(t) = p^r q^r e^t (1 - qe^t)^{-r-1} + p^r q^r e^t (-r-1) (1 - qe^t)^{-r-2} \\ (-qe^t) + p^r q^2 r(r+1) 2e^{2t} (1 - qe^t)^{-r-2} + p^r q^2 r \\ (r+1) e^{2t} (-r-2) (1 - qe^t)^{-r-3} (-qe^t) + 2p^r q^2 r \\ (r+1) 2e^{2t} (1 - qe^t)^{-r-2} + 2p^r q^2 r(r+1) e^{2t} (r+2) \\ (1 - qe^t)^{-r-3} (-qe^t) + p^r q^3 r(r+1)(r+2) 3e^{3t} \\ (1 - qe^t)^{-r-3} + p^r q^3 r(r+1)(r+2) e^{3t} (-r-3) \\ (1 - qe^t)^{-r-4} (-qe^t)$$

$$= p^r q^r e^t (1 - qe^t)^{-r-1} + p^r q^2 r(r+1) e^{2t} (1 - qe^t)^{-r-2} + \\ p^r q^2 r(r+1) 2e^{2t} (1 - qe^t)^{-r-2} + p^r q^3 r(r+1) e^{3t} \\ (1 - qe^t)^{-r-3} + 2p^r q^2 r(r+1) 2e^{2t} (1 - qe^t)^{-r-2} + \\ 2p^r q^3 r(r+1)(r+2) e^{3t} (1 - qe^t)^{-r-3} + p^r q^3 r(r+1) \\ (r+2) 3e^{3t} (1 - qe^t)^{-r-3} + p^r q^4 r(r+1)(r+2)(r+3) \\ e^{4t} (1 - qe^t)^{-r-4}$$

put $t=0$

$$= p^r q^r (1 - q)^{-r-1} + p^r q^2 r(r+1) (1 - q)^{-r-2} + 2p^r q^2 r \\ (r+1) (1 - q)^{-r-2} + p^r q^3 r(r+1) (1 - q)^{-r-3} + 4p^r q^3 \\ r(r+1) (1 - qe^t)^{-r-2} + 2p^r q^3 r(r+1)(r+2) (1 - q)^{-r-3} \\ + 3p^r q^3 r(r+1)(r+2) (1 - q)^{-r-3} + p^r q^4 r(r+1)(r+2)$$

$$(r+3)(1-q)^{-r-4}$$

$$\begin{aligned}
 &= p^3 q^2 (p^{-r-1}) + p^r q^2 r(r+1)(p)^{-r-2} + 2p^2 q^2 r(r+1) \\
 &\quad (p^{-r-2}) + p^r q^3 r(r+1)(p)^{-r-3} + 4p^1 q^3 r(r+1)(p^{-r-2}) \\
 &\quad + 2p^1 q^3 r(r+1)(r+2)(p^{-r-3}) + 3p^r q^3 r(r+1)(r+2) \\
 &\quad (p^{-r-3}) + p^r q^4 r(r+1)(r+2)(r+3)(p^{-r-4}) \\
 &= \frac{p}{r} r + \frac{q^2}{r(r+1)} + 2 \frac{q^2}{r(r+1)} + \frac{q^3}{r(r+1)} + \frac{q^3}{r(r+1)} + \frac{q^3}{r(r+1)} + \frac{q^3}{r(r+1)} \\
 &\quad + 2 \frac{q^3}{r(r+1)} + 3 \frac{q^3}{r(r+1)} + \frac{q^4}{r(r+1)} + \frac{q^4}{r(r+1)} + \frac{q^4}{r(r+1)} + \frac{q^4}{r(r+1)} \\
 &\quad (r+2)(r+3)
 \end{aligned}$$

$$E(x^n) = \frac{q}{p} r + 7 \frac{q^2}{r(r+1)} + 6 \frac{q^3}{r(r+1)(r+2)} + \frac{q^4}{r(r+1)(r+2)(r+3)}$$

Negative Binomial Distribution

Factorial G.F. of Negative Binomial Distribution

$$H_0(\alpha) = p^r (p - q\alpha)^{-r}$$

By taking log of F.G.F we get.

$$\begin{aligned} H_0(\alpha) &= \log(p^r (p - q\alpha)^{-r}) \\ &= \log(p^r) + \log((p - q\alpha)^{-r}) \\ &= r \log p + (-r) \log(p - q\alpha) \\ &= r \log p - r \log(p - q\alpha) \end{aligned}$$

Now Taking Derivative w.r.t α - -

1st -

$$\begin{aligned} h_0(\alpha) &= 0 + (-r)(p - q\alpha)^{-1}(-q) \\ &= qr(p - q\alpha)^{-1} \end{aligned}$$

$$\text{put } \alpha = 0$$

$$= qr(p - 0)^{-1}$$

$$= qr p^{-1}$$

$$K_0(1) = qr/p$$

2nd:-

$$q^2 r (p - q\alpha)^{-1}$$

$$h_0''(\alpha) = q^2 r (-1) (p - q\alpha)^{-2} (-q)$$

$$= q^3 r (p - q\alpha)^{-2}$$

$$\text{put } \alpha = 0$$

$$= q^3 r (p - 0)^{-2}$$

$$= q^3 r p^{-2}$$

$$= q^3 r / p^2$$

3rd:-

$$q^3 r (p - q\alpha)^{-2}$$

$$h_0'''(\alpha) = q^3 r (-2) (p - q\alpha)^{-3} (-q)$$

$$= 2q^4 r (p - q\alpha)^{-3}$$

$$\text{put } \alpha = 0$$

$$= 2q^4 r (p - 0)^{-3}$$

$$= 2q^4 r p^{-3}$$

$$= 2q^4 r / p^3$$

4th:-

$$2q^4 r (p - q\alpha)^{-3}$$

$$h_0^{iv}(\alpha) = 2q^4 r (-3) (p - q\alpha)^{-4} (-q)$$

$$= 6q^5 r (p - q\alpha)^{-4}$$

$$\text{put } \alpha = 0$$

$$6q^4r(p-0)^{-1}$$

$$= 6q^4r p^{-1}$$

$$= 6q^4r/p$$

Comulents (A'apais)

1st:-

$$K_1 = K_0(1) = m'$$

$$= q^2/p$$

2nd:-

$$K_2 = K_0(1) + K_0(2) = 11$$

$$= q^2/p + q^2/p^2$$

$$= \frac{pq^2 + q^3}{p^2}$$

$$= \frac{q^2(p+q)}{p^2}$$

$$= \frac{q^2}{p^2}$$

$$\therefore (p+q) = 1$$

3rd:-

$$K_3 = K_0(1) + 3K_0(2) + K_0(3) = \mu_3$$

$$= q^2/p + 3q^2/p^2 + 2q^3/p^3$$

$$= \frac{q^2}{p} \left[1 + \frac{3q}{p} + \frac{2q^2}{p^2} \right]$$

$$= \frac{q^2}{p} \left[\frac{p^2 + 3pq + 2q^2}{p^2} \right]$$

$$= \frac{q^2}{p^3} [p^2 + q^2 + 2pq + q^2 + pq]$$

$$= \frac{q^2}{p^3} [(p+q)^2 + q(q+p)]$$

$$= \frac{q^2}{p^3} [(1)^2 + q(1)]$$

$$= \frac{q^2}{p^3} (1+q)$$

4th:-

$$K_4 = K_0(1) + 7K_0(2) + 6K_0(3) + K_0(4)$$

$$= q^2/p + 7q^2/p^2 + 6(2q^3/p^3) + 6q^4/p^4$$

$$= q^2/p + 7q^2/p^2 + 12q^3/p^3 + 6q^4/p^4$$

$$= q^2/p \left[1 + 7q/p + 12q^2/p^2 + 6q^3/p^3 \right]$$

$$= \frac{q_1}{p} \left(\frac{p^3 + 7p^2q + 13pq^2 + 6q^3}{p^3} \right)$$

$$= \frac{q_1}{p^4} (q^3 + 3p^2q + 3p^2q + p^3 + 4p^2q + 9pq^2 + 5q^3)$$

$$= \frac{q_1}{p^4} ((q+p)^3 + 4p^2q + 9pq^2 + 5q^3)$$

$$K_4 = \frac{q_1}{p^4} (1 + 4p^2q + 9pq^2 + 5q^3)$$

$$L_4 = K_4 + 3(K_2)^2$$

$$= \frac{q_1}{p^4} (1 + 4p^2q + 9pq^2 + 5q^3) + 3 \frac{q_1^2}{p^4}$$

$$= \frac{q_1}{p^4} (1 + 4p^2q + 9pq^2 + 5q^3 + 3q_1)$$

$$= \frac{q_1}{p^4} (1 + q(4p^2 + 9pq + 5q^2) + 3q_1)$$

$$= \frac{q_1}{p^4} (1 + q(2p^2 + 2q^2 + 8pq + pq + q^2) + 3q_1)$$

$$= \frac{q_1}{p^4} (1 + q(2p + 2q)^2 + pq + q^2 + 3q_1)$$

$$= \frac{q_1}{p^4} (1 + q(4(p+q)^2 + q(p+q) + 3q_1))$$

$$\mu_4 = \frac{qr}{p^4} [1 + 4q(1)^2 + q^2 + 3qr]$$

$$= \frac{qr}{p^4} [1 + 4q + q^2 + 3qr]$$

Skewness

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

$$= \frac{qr/p^3(1+q)}{(qr/p^2)^{3/2}}$$

$$= \frac{qr/p^3(1+q)}{qr/p^3 \cdot \frac{\sqrt{qr}}{p}}$$

$$= \frac{(1+q)/p}{\sqrt{qr}/p}$$

$$\text{Skewness} = \frac{1+q}{\sqrt{qr}}$$

KL Divergence

$$KL Divergence = \frac{L_1}{L_2}$$

$$= \frac{9 \times 10^4 [1 + 49 + 9^2 + 396]}{9 \times 10^4}$$

$$KL Divergence = \frac{1 + 49 + 9^2 + 396}{9}$$

$$= \frac{455}{9}$$

Cumulant Generating Function

Negative Binomial Distribution

Moment Generating Function of
Negative Binomial Distribution :

$$M_0(t) = p^r (1 - qe^t)^{-r}$$

By taking log of M.G.F we get.

$$\begin{aligned} C(t) &= \log(p^r (1 - qe^t)^{-r}) \\ &= rp + (-r)(1 - qe^t) \\ &= rp - r(1 - qe^t) \end{aligned}$$

1st:-

$$\begin{aligned} C'_0(t) &= 0 - r(1 - qe^t)^{-1}(-qe^t) \\ &= rqe^t(1 - qe^t)^{-1} \end{aligned}$$

$$\text{put } t=0$$

$$= rq(1 - q)^{-1}$$

$$= rq(p)^{-1}$$

$$= rq/p$$

2nd:

$$rqe^t(1-qe^t)^{-1}$$

$$\begin{aligned}C''(t) &= rq[e^t(1-qe^t)^{-1} + e^t(-1)(1-qe^t)^{-2}(-qe^t)] \\&= rq[e^t(1-qe^t)^{-1} + qe^{2t}(1-qe^t)^{-2}] \\&= rqe^t(1-qe^t)^{-2}[(1-qe^t) + qe^t]\end{aligned}$$

put $t=0$

$$\begin{aligned}&= rq(1-q)^{-2}[(1-q) + q] \\&= rq(p)^{-2}(p+q) \\&= rq/p^2(1) \\&= rq/p^2\end{aligned}$$

3rd:

$$\begin{aligned}C'''(t) &= rq[e^t(1-qe^t)^{-1} + rq^2e^{2t}(1-qe^t)^{-2}] \\&= rq[e^t(1-qe^t)^{-1} + e^t(-1)(1-qe^t)^{-2}(-qe^t)] \\&\quad + rq^2[2e^{2t}(1-qe^t)^{-2} + e^{2t}(-2)(1-qe^t)^{-3}(-qe^t) - qe^t] \\&= rq[e^t(1-qe^t)^{-1} + qe^{2t}(1-qe^t)^{-2}] + \\&\quad rq^2[2e^{2t}(1-qe^t)^{-2} + 2qe^{3t}(1-qe^t)^{-3}] \\&= rqe^t(1-qe^t)^{-2}[(1-qe^t) + qe^t] + \\&\quad 2rq^2e^{2t}(1-qe^t)^{-3}[(1-qe^t) + qe^t]\end{aligned}$$

$$\text{put } t = 0$$

$$= r q (1-q)^{-2} [(1-q)+q] + 2 r q^2 (1-q)^{-3} [(1-q)+q]$$

$$= r q (p)^{-2} (q+p) + 2 r q^2 (p)^{-3} (q+p)$$

$$= r q / p^2 (1) + 2 r q^2 / p^3 (1)$$

$$= r q / p^2 + 2 r q^2 / p^3$$

4th:

$$r q e^t (1-q e^t)^{-1} + r q^2 e^{2t} (1-q e^t)^{-2} + 2 r q^2 e^{2t} (1-q e^t)^{-2} + 2 r q^3 e^{3t} (1-q e^t)^{-3}$$

$$\begin{aligned} C^{iv}(t) &= r q [e^t (1-q e^t)^{-1} + e^t (-1) (1-q e^t)^{-2} (-q e^t)] \\ &\quad + r q^2 [2 e^{2t} (1-q e^t)^{-2} + e^{2t} (-2) (1-q e^t)^{-3} (-q e^t)] \\ &\quad + 2 r q^2 [2 e^{2t} (1-q e^t)^{-2} + e^{2t} (-2) (1-q e^t)^{-3} (-q e^t)] \\ &\quad + 2 r q^3 [3 e^{3t} (1-q e^t)^{-3} + e^{3t} (-3) (1-q e^t)^{-4} (-q e^t)] \\ &= r q e^t (1-q e^t)^{-2} [(1-q e^t) + q e^t] + r q^2 2 e^{2t} (1-q e^t)^{-3} [(1-q e^t) + q e^t] \\ &\quad + 2 r q^2 2 e^{2t} (1-q e^t)^{-3} [(1-q e^t) + q e^t] + 2 r q^3 3 e^{3t} (1-q e^t)^{-4} [(1-q e^t) + q e^t] \end{aligned}$$

$$\text{put } t=0$$

$$\begin{aligned} &= r q (1-q)^{-2} [q+p] + 2 r q^2 (1-q)^{-3} [q+p] \\ &\quad + 4 r q^2 (1-q)^{-3} [q+p] + 6 r q^3 (1-q)^{-4} [q+p] \end{aligned}$$

$$\begin{aligned}
 &= xq/p^2 + 2xq^2/p^3 + 4xq^2/p^3 + 6xq^3/p^4 \\
 &= xq/p^2 + 6xq^2/p^3 + 6xq^3/p^4.
 \end{aligned}$$

$$\begin{aligned}
 K_q &= K_q + 3K_2 \\
 &= \frac{q}{p^2} x + \frac{6q^2}{p^3} x + \frac{6q^3}{p^4} x + \frac{3q^2}{p^4} x
 \end{aligned}$$

Factorial Negative Binomial Function Distribution.

$$M_0(t) = p^r (1 - qe^t)^{-r}$$

By replacing (e^t) by (α) we get.

$$M_0(\alpha) = p^r (1 - q\alpha)^{-r}$$

By taking derivative we get.

1st:-

$$H'_0(t) = p^r (-r) (1 - q\alpha)^{-r-1} (-q\alpha)$$

$$= p^r q r \alpha (1 - q\alpha)^{-r-1}$$

$$\text{put } \alpha = 0$$

$$= p^r q r (1 - q)^{-r-1}$$

$$= p^r q r (p^{-r-1})$$

$$E(x) = \frac{qr}{p}$$

2nd:-

$$p^r q r \alpha (1 - q\alpha)^{-r-1}$$

$$H''_0(t) = p^r q r (-r-1) (1 - q\alpha)^{-r-2} (-q)$$

$$= p^r q^2 r (r+1) (1 - q\alpha)^{-r-2}$$

$$\text{put } \alpha = 0$$

$$= p^r q^2 r(r+1) (1-q)^{-r-2}$$

$$= p^r q^2 r(r+1) (p^{-r-2})$$

$$\text{Ans} = \frac{q^2}{p^2} r(r+1) = E[x(x-1)]$$

3rd:-

$$p^r q^2 r(r+1) (1-q\alpha)^{-r-2}$$

$$H_0'''(t) = p^r q^2 r(r+1) (-r-2) (1-q\alpha)^{-r-3} (-q)$$

$$= p^r q^3 r(r+1) (r+2) (1-q\alpha)^{-r-3}$$

$$\text{put } \alpha = 0$$

$$= p^r q^3 r(r+1) (r+2) (1-q)^{-r-3}$$

$$= p^r q^3 r(r+1) (r+2) (p^{-r-3})$$

$$\text{Ans} = \frac{q^3}{p^3} r(r+1) (r+2) = E[x(x-1)(x-2)]$$

4th:-

$$p^r q^3 r(r+1) (r+2) (1-q\alpha)^{-r-3}$$

$$H_0^{iv}(t) = p^r q^3 r(r+1) (r+2) (-r-3) (1-q\alpha)^{-r-4} (-q)$$

$$= p^r q^4 r(r+1) (r+2) (r+3) (1-q\alpha)^{-r-4}$$

$$\text{put } \alpha = 0$$

$$= p^r q^4 r(r+1) (r+2) (r+3) (1-q)^{-r-4}$$

$$= \frac{q^4}{p^4} \tau(\tau+1)(\tau+2)(\tau+3)(p-\tau-4)$$

$$= \frac{q^4}{p^4} \tau(\tau+1)(\tau+2)(\tau+3) + E[\tau(\tau-1)(\tau-2)(\tau-3)]$$

Ordinary Moments (OR)
Moments About Origin.

$$\mu'_1 = E(X) = \frac{q\tau}{p}$$

$$\mu'_2 = E[\tau(\tau-1)] + E(X) = \frac{q^2}{p^2} \tau(\tau+1) + \frac{q\tau}{p}$$

$$\mu'_3 = E[\tau(\tau-1)(\tau-2)] + 3E\left[\frac{p^2}{\tau(\tau-1)}\right] + E(X)$$

$$= \frac{q^3}{p^3} \tau(\tau+1)(\tau+2) + 3 \frac{q^2}{p^2} \tau(\tau+1) + \frac{q\tau}{p}$$

$$\mu'_4 = E[\tau(\tau-1)(\tau-2)(\tau-3)] + 6E[\tau(\tau-1)(\tau-2)]$$

$$+ 7E[\tau(\tau-1)] + E(X)$$

$$= \frac{q^4}{p^4} \tau(\tau+1)(\tau+2)(\tau+3) + 6 \frac{q^3}{p^3} \tau(\tau+1)(\tau+2)$$

$$+ 7 \frac{q^2}{p^2} \tau(\tau+1) + \frac{q\tau}{p}$$

Moments About Mean

1st Moment:

$$m_1 = m'_1 - m'_1 = 0$$

$$\begin{aligned} &= \frac{x_1}{p} - \frac{x_1}{p} \\ &= 0 \end{aligned}$$

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2nd Moment:

$$m_2 = m'_2 - (m'_1)^2$$

$$= \gamma(\gamma+1) \frac{q^2}{p^2} + \frac{\gamma q}{p} - \left(\frac{\gamma q}{p} \right)^2$$

$$= \gamma(\gamma+1) \frac{q^2}{p^2} + \frac{\gamma q}{p} - \frac{\gamma^2 q^2}{p^2}$$

$$= \frac{\gamma^2 q^2}{p^2} + \frac{\gamma q^2}{p^2} + \frac{\gamma q}{p} - \frac{\gamma^2 q^2}{p^2}$$

$$= \frac{\gamma^2 q^2}{p^2} - \frac{\gamma^2 q^2}{p^2} + \frac{\gamma q^2}{p^2} + \frac{\gamma q}{p}$$

$$= \frac{\gamma q^2}{p^2} + \frac{\gamma q}{p}$$

$$= \frac{\gamma q}{p} \left[\frac{q}{p} + 1 \right]$$

$$= \frac{\gamma q}{p} \left[\frac{q+p}{p} \right]$$

$$= \frac{\gamma q}{p} \left[\frac{1}{p} \right]$$

as $q+p=1$

$$m_2 = \frac{\gamma q}{p^2}$$

3rd Moment:

$$m_3 = m_1' - 3m_2'm_1' + 2(m_1')^3$$

$$= \gamma(\gamma+1)(\gamma+2)\frac{q^3}{p^3} + 3\gamma(\gamma+1)\frac{q^2}{p^2} + \frac{\gamma q}{p} - 3$$

$$+ \left[\left(\gamma(\gamma+1)\frac{q^2}{p^2} \right) + \left(\frac{\gamma q}{p} \right) \right] \left[\frac{\gamma q}{p} \right] + \frac{2\gamma^2 q^3}{p^3}$$

$$= \gamma(\gamma+1)(\gamma+2)\frac{q^3}{p^3} + 3\gamma(\gamma+1)\frac{q^2}{p^2} + \frac{\gamma q}{p}$$

$$+ \frac{2\gamma^2(\gamma+1)q^3}{p^3} + \frac{\gamma^2 q^2}{p^2} + \frac{2\gamma^2 q^3}{p^3}$$

$$+ \frac{\gamma q}{p} \left[(\gamma+1)(\gamma+2)\frac{q^2}{p^2} + 3(\gamma+1)\frac{q}{p} + 1 \right]$$

$$= 3\gamma(\gamma+1)\frac{q^2}{p^2} + \frac{23\gamma q}{p} + \frac{2\gamma^2 q^3}{p^3}$$

$$= \frac{\gamma q}{p} \left[\frac{3(\gamma+1)(\gamma+2)q^2 + 3(\gamma+1)pq + p^2 - 3\gamma(\gamma+1)}{p^2} - \frac{3pq + 2q^2}{p^2} \right]$$

$$= \frac{\gamma q}{p^3} \left[(\gamma+1)(\gamma+2)q^2 + 3(\gamma+1)pq + p^2 - 3\gamma(\gamma+1) - 3pq + 2q^2 \right]$$

$$= \frac{\gamma q}{p^3} \left[q^2\gamma + 3q^2 + 3p^2\gamma + 3pq + p^2 - 3q^2 - 3pq \right]$$

$$= \frac{r q}{p^3} \left[p^3 + 2q^2 + 3pq \right]$$

$$= \frac{r q}{p^3} \left[p^3 + q^2 + 2pq + q^2 + pq \right]$$

$$= \frac{r q}{p^3} \left[(p+q)^2 + q(q+p) \right]$$

$$= \frac{r q}{p^3} \left[1 + q \right]$$

$$m_3 = \frac{r q}{p^3} (1+q)$$

4th Moment:

$$m_4 = m_4' - 4m_3' m_1' + 6m_2' (m_1')^2 - 3(m_1')^4$$

$$m_4 = r(r+1)(r+2)(r+3) \frac{q^4}{p^4} + 6r(r+1)(r+2) \frac{q^3}{p^3} + 7r(r+1) \frac{q^2}{p^2} + \frac{r q}{p}$$

$$- 4 \left[\left(r(r+1)(r+2) \frac{q^3}{p^3} + 3r(r+1) \frac{q^2}{p^2} + \frac{r q}{p} \right) \left(\frac{r q}{p} \right) \right] + 6$$

$$\left[\left(r(r+1) \frac{q^2}{p^2} + \frac{r q}{p} \right) \left(\frac{r^2 q^2}{p^2} \right) \right] - 3 \left[\frac{r^4 q^4}{p^4} \right]$$

$$\begin{aligned}
 & x(x+1)(x+2)(x+3) \frac{q''}{p^4} + 6x(x+1)(x+2) \frac{q'''}{p^3} + 7x(x+1) \frac{q^{(4)}}{p^2} \\
 & + \frac{xq}{p} - 4 \left[\frac{x^2 q''}{p^4} (x^2 + 3x + 2) + \frac{3x^2 q'''}{p^3} (x+1) + \frac{x^2 q^{(4)}}{p^2} \right] + 6 \\
 & \left[\frac{x^2 q''}{p^4} (x+1) + \frac{x^2 q'''}{p^3} \right] - 3 \frac{x^2 q^{(4)}}{p^4}
 \end{aligned}$$

$$\frac{yq}{p} \left[\begin{aligned}
 & x(x+1)(x+2)(x+3) \frac{q''}{p^3} + 6(x+1)(x+2) \frac{q'''}{p^2} + 7(x+1) \frac{q^{(4)}}{p} + 1 \\
 & - 11 \frac{q''}{p^3} (x^2 + 3x + 2) - 12 \frac{xq'''}{p^2} (x+1) - 4 \frac{xq}{p} + 6 \frac{x^2 q'''}{p^2} (x+1) + 6 \frac{x^2 q^{(4)}}{p^2} - 3 \frac{x^2 q^{(4)}}{p^2}
 \end{aligned} \right]$$

$$= \frac{yq}{p} \left[\frac{\begin{aligned}
 & (x^3 + 6x^2 + 11x + 6)q'' + 6(x^2 + 3x + 2)pq''' + 7(x+1)p^2q^{(4)} + p^3 - 4q'' \\
 & (x^2 + 3x + 2) - 12pq'''(x+1) - 4p^2q'' + 6q''^2(x+1) + 6pq''^2 - 3q''^2
 \end{aligned}}{p^3} \right]$$

$$= \frac{yq}{p^4} \left[\begin{aligned}
 & q''^3/p^3 + 6q''^2/p^2 + 11q''^2/p + 6q''^2 + 6pq''^2/p + 18pq''^2/p + 12pq''^2 + 7p^2q'' + 7p^2q'' \\
 & + p^3 - 4q''^3/p - 8q''^3/p - 12pq''^2/p - 12pq''^2/p - 4p^2q'' + 6q''^2/p + 6q''^2/p + \\
 & 6pq''^2/p - 3q''^3/p - 12q''^3/p
 \end{aligned} \right]$$

$$= \frac{yq}{p^4} \left[\begin{aligned}
 & 11q''^2/p + 6q''^2 + 18pq''^2/p + 12pq''^2 + 7p^2q'' + 7p^2q'' + p^3 \\
 & - 8q''^3/p - 12pq''^2/p - 4p^2q''
 \end{aligned} \right]$$

$$= \frac{yq}{p^4} \left[\begin{aligned}
 & 3q''^2/p + 6q''^2 + 6pq''^2/p + 12pq''^2 + 8p^2q'' + 7p^2q'' + p^3
 \end{aligned} \right]$$

$$= \frac{r_9}{r^9} \left[r^3 + q^3 + 3r^2q + 3rq^2 + 5q^3 + 5r^2q + 6rq^2 + 3r^2q \right]$$

$$= \frac{r_9}{r^9} \left[(r+q)^3 + q(4r^2 + 9rq + 5q^2) + 3qr + (r^2 + 2rq + q^2) \right]$$

$$= \frac{r_9}{r^9} \left[1 + q \left[(2r+2q)^2 + 4rq + q^2 \right] + 3qr \right] \quad \text{As } r+q=1$$

$$= \frac{r_9}{r^9} \left[1 + q \left[4(r+q)^2 + 4rq + q^2 \right] + 3qr \right]$$

$$= \frac{r_9}{r^9} \left[1 + q \left[4(1)^2 + 4q + q^2 \right] + 3qr \right]$$

$$= \frac{r_9}{r^9} \left[1 + q \left[4 + 4q + q^2 \right] + 3qr \right]$$

$$= \frac{r_9}{r^9} \left[1 + 4q + q^2 + 3qr \right]$$

So,

$$m_4 = \frac{r_9}{r^9} \left[1 + 4q + q^2 + 3qr \right]$$

Log Normal Distribution

Moments About Origin

1st Moment :-

$$\begin{aligned} E(X) &= \int x f(x) \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_0^{\infty} \frac{x}{x} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} dx \end{aligned}$$

put : $\frac{\log x - \mu}{\sigma} = y$ we get.

and : $\log x - \mu = \sigma y$ $0 < x < \infty$
 $-\infty < y < \infty$

$$\frac{1}{x} dx = \delta \ln y \quad , \quad x = e^{\delta y + \mu}$$

$$= \frac{1}{\sqrt{2\pi}\delta} \int_{-\infty}^{\infty} e^{\delta y + \mu} \cdot e^{-\frac{1}{2}\delta^2 y^2} \cdot \delta dy$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \delta \int_{-\infty}^{\infty} e^{\delta y + \mu - \frac{1}{2}\delta^2 y^2} dy$$

$$= \frac{e^{\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2\delta y + \delta^2 - \delta^2)} dy$$

$$= e^{\mu + \frac{1}{2}\delta^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - \delta)^2} dy$$

$$\text{let } y - \delta = t, \quad dy = dt$$

$$-\infty < y < \infty, \quad -\infty < t < \infty$$

$$= e^{\mu + \frac{1}{2}\delta^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}t^2} dt$$

$$E(x) = e^{\mu + \frac{1}{2}\delta^2}$$

2nd Moment:

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \frac{1}{\sqrt{2\pi}\delta} \int_0^{\infty} \frac{x^2}{x} \cdot e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\delta}\right)^2} dx$$

$$\text{put } \frac{\log x - \mu}{\delta} = y, \quad x = e^{\delta y + \mu}$$

$$\log x - \mu = \delta y, \quad \frac{1}{x} dx = \delta dy$$

$$0 < x < \infty, \quad -\infty < y < \infty$$

$$= \frac{1}{\sqrt{2\pi}\delta} \int_{-\infty}^{\infty} (e^{\delta y + \mu})^2 \cdot e^{-\frac{1}{2}y^2} \cdot \delta dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{2\delta y + 2\mu - \frac{1}{2}y^2} dy$$

$$= \frac{e^{2\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 4\delta y + 4\delta^2 - 4\delta^2)} dy$$

$$= e^{2\mu + 2\delta^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - 2\delta)^2} dy$$

Let: $y - 2\sigma = t$, $dy = dt$
 $-\infty < y < \infty$, $-\infty < t < \infty$

$$E(X^2) = e^{\frac{2\mu + 2\sigma^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}t^2} dt$$

$$E(X^2) = e^{\frac{2\mu + 2\sigma^2}{2}} \quad t \sim N(0, 1)$$

Variance:-

$$\begin{aligned} V(X) &= E(X^2) - (E(X))^2 \\ &= e^{\frac{2\mu + 2\sigma^2}{2}} - \left[e^{\frac{\mu + 1/2\sigma^2}{2}} \right]^2 \\ &= e^{\frac{2\mu + 2\sigma^2}{2}} - e^{\frac{2\mu + \sigma^2}{2}} \\ &= e^{\frac{2\mu + \sigma^2}{2}} [e^{\frac{\sigma^2}{2}} - 1] \end{aligned}$$

3rd Moment:-

$$E(X^3) = \int_{-\infty}^{\infty} x^3 f(x) dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \frac{x^3}{x} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} dx$$

$$\text{let } \frac{\log x - \mu}{\delta} = y \quad , \quad x = e^{\delta y + \mu}$$

$$\log x - \mu = \delta y \quad , \quad \frac{1}{x} dx = \delta dy$$

$$0 < x < \infty \quad , \quad -\infty < y < \infty$$

$$E(x^3) = \frac{1}{\sqrt{2\pi} \delta} \int_{-\infty}^{\infty} (e^{\delta y + \mu})^3 \cdot e^{-\frac{1}{2}y^2} \delta dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{3\delta y + 3\mu - \frac{1}{2}y^2} dy$$

$$= \frac{e^{3\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 6\delta y + 9\delta^2 + 9\delta^2)} dy$$

$$= e^{3\mu + 9\delta^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - 3\delta)^2} dy$$

$$\text{let } : y - 3\delta = t \quad , \quad dy = dt$$

$$-\infty < y < \infty \quad , \quad -\infty < t < \infty$$

$$= e^{\frac{3\mu + 9\sigma^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{1}{2}t^2} dt$$

$$E(x^3) = e^{\frac{3\mu + 9\sigma^2}{2}}$$

4th Moment:-

$$E(x^4) = \int_{-\infty}^{\infty} x^4 f(x) dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \frac{x^4}{x} \cdot e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} dx$$

Let: $\frac{\log x - \mu}{\sigma} = y$, $x = e^{\sigma y + \mu}$

$$\log x - \mu = \sigma y, \quad \frac{1}{x} dx = \sigma dy$$

$$0 < x < \infty \quad \rightarrow \quad -\infty < y < \infty$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (e^{\sigma y + \mu})^4 \cdot e^{-\frac{1}{2}y^2} \cdot \sigma dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{4\delta y + 4\mu - 1/2 y^2} dy$$

$$= \frac{e^{4\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 (y^2 - 8\delta y + 16\delta^2 - 16\delta^2)} dy$$

$$= \frac{e^{4\mu} e^{16\delta^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 (y - 4\delta)^2} dy$$

let: $y - 4\delta = t$, $dy = dt$

$-\infty < y < \infty$, $-\infty < t < \infty$

$$= e^{4\mu + 16\delta^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 t^2} dy$$

$\phi(y) = e^{4\mu + 16\delta^2}$

Moments About Mean

1st Moment:-

$$\begin{aligned}\mu_1 &= \mu'_1 - \mu'_1 = 0 \\ &= e^{\mu + 1/2 \sigma^2} - e^{\mu + 1/2 \sigma^2} \\ &= 0\end{aligned}$$

2nd Moment:-

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= e^{2\mu + 2\sigma^2} - (e^{\mu + 1/2 \sigma^2})^2 \\ &= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} \\ &= e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)\end{aligned}$$

3rd Moment:-

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_1 \mu'_1 + 2(\mu'_1)^3 \\ &= e^{3\mu + 9\sigma^2} - 3(e^{2\mu + 2\sigma^2})(e^{\mu + 1/2 \sigma^2}) + 2e^{3\mu + 3/2 \sigma^2} \\ &= e^{3\mu + 9\sigma^2} - 3e^{2\mu + \mu + 2\sigma^2 + 1/2 \sigma^2} + 2e^{3\mu + 3/2 \sigma^2} \\ &= e^{3\mu + 9\sigma^2} - 3e^{3\mu + 5/2 \sigma^2} + 2e^{3\mu + 3/2 \sigma^2} \\ &= e^{3\mu + 3/2 \sigma^2} [e^{4/2 \sigma^2} - 3e^{1/2 \sigma^2} + 2]\end{aligned}$$

4th Moment :-

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_2)^2 - 3(\mu'_1)^4$$

$$= e^{4\mu + 16\delta^2} - 4 \left[(e^{3\mu + 9\delta^2})(e^{\mu + 1/4\delta^2}) \right] +$$

$$6 \left[(e^{2\mu + 2\delta^2})(e^{2\mu + 8\delta^2}) \right] - 3e^{4\mu + 8\delta^2}$$

$$= e^{4\mu + 16\delta^2} - 4 \left(e^{4\mu + 1/2\delta^2} \right) + 6 \left(e^{4\mu + 3\delta^2} \right)$$

$$- 3e^{4\mu + 2\delta^2}$$

$$= e^{4\mu + 2\delta^2} \left[e^{14\delta^2} - 4e^{1/2\delta^2} + 6e^{3\delta^2} - 3 \right]$$

Median of Lognormal Distribution

$$\text{Median} = \int f(x) dx = \frac{1}{2}$$

$$\bullet \frac{1}{\sqrt{2\pi}\delta} \int_0^M \frac{1}{x} \cdot e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\delta}\right)^2} dx = \frac{1}{2}$$

$$\text{put } \frac{\log x - \mu}{\delta} = y, \quad x = e^{\delta y + \mu}$$

$$\log x - \mu = \delta y \Rightarrow \frac{1}{x} dx = \delta dy$$

$$0 < x < M \Rightarrow -\infty < y < \frac{\log M - \mu}{\delta}$$

$$\frac{1}{\sqrt{2\pi}\delta} \int_{-\infty}^{\log M - \mu / \delta} e^{-\frac{1}{2}y^2} \cdot \delta dy = \frac{1}{2}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\log M - \mu / \delta} e^{-\frac{1}{2}y^2} dy = \frac{1}{2}$$

$$y \sim N(0, 1)$$

$$\Phi\left(\frac{\log M - \mu}{\delta}\right) = \frac{1}{2}$$

$$\frac{\log M - \mu}{\delta} = 0$$

$$\log M = \mu$$

$$N_{\text{rad}} = e^{\mu}$$

Mode of LogNormal Distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} \cdot e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2}$$

differentiate w.r.t 'x'

$$\log f(x) = \log\left(\frac{1}{\sqrt{2\pi}\sigma x}\right) + \log\left(\frac{1}{x}\right) - \frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2$$

$$\log f(x) = f' \cdot \frac{1}{f}$$

$$\frac{f'}{f} = 0 + x\left(-\frac{1}{x^2}\right) - \frac{2}{2\sigma^2}(\log x - \mu) \cdot \frac{1}{x}$$

$$= -\frac{1}{x} - \frac{1}{\sigma^2 x}(\log x - \mu)$$

$$= -\frac{1}{x} \left[1 + \frac{1}{\sigma^2}(\log x - \mu) \right]$$

$$f' = -\frac{1}{x} \left[1 + \frac{1}{\sigma^2}(\log x - \mu) \right] \cdot f$$

$$\therefore -\frac{1}{x} \left[1 + \frac{1}{\sigma^2}(\log x - \mu) \right] \cdot f = 0$$

$$-1 = \frac{1}{x} \frac{1}{x\delta^2} (\log x - \mu)$$

$$\frac{1}{\delta^2} (\log x - \mu) = -1$$

$$\log x - \mu = -\delta^2$$

$$\log x = \mu - \delta^2$$

$$x = e^{\mu - \delta^2}$$

$$f'(x) < 0$$

$$Mode = e^{\mu - \delta^2}$$

PUACF

Geometric Distribution is Moments About Origin

1st Moment:-

$$E(x) = \sum_{x=0}^{\infty} x \cdot p(x)$$

$$= \sum_{x=0}^{\infty} x \cdot p \cdot q^x$$

$$= 0 + (1) pq + (2) pq^2 + 3pq^3 + \dots$$

$$= pq + 2pq^2 + 3pq^3 + \dots$$

$$= pq(1 + 2q + 3q^2 + \dots)$$

$$= pq(1 - q)^{-2}$$

$$= pq(p)^{-2}$$

As: $1 - q = p$

$$= p \cdot q \cdot p^{-2}$$

$$= p^{-1} \cdot q$$

$$= p^{-1} \cdot q$$

$$E(x) = q/p$$

2nd: Moment: $E(x^2)$

3rd: Moment: $E(x^3)$

$$E(x^2) = E[x(x-1)] + E(x) \quad \text{--- (A)}$$

$$E[x(x-1)] = \sum_{x=0}^{\infty} x(x-1) pq^x$$

$$= 0 + 0 + (2) pq^2 + 3(2) pq^3 + 4(3) pq^4 + \dots$$

$$= 2pq^2 + 6pq^3 + 12pq^4 + \dots$$

$$= 2pq^2 [1 + 3q + 6q^2 + \dots]$$

$$= 2pq^2 (1-q)^{-3}$$

$$= 2pq^2 (p)^{-3}$$

As: $1-q=p$

$$= 2pq^2 \cdot p^{-3}$$

$$= 2p^{-2}q^2$$

$$E[x(x-1)] = \frac{2q^2}{p^2}$$

$$E(x^2) = E[x(x-1)] + E(x) \quad \text{--- (A)}$$

$$E(x^2) = \frac{2q^2}{p^2} + \frac{q}{p} = \frac{q}{p} \left[\frac{2q}{p} + 1 \right]$$

3rd Moment:-

$$E(X^3) = E[X(X-1)(X-2)] + 3E[X(X-1)] + E(X) \quad \text{--- (1)}$$

$$E[X(X-1)(X-2)] = \sum_{x=0}^{\infty} x(x-1)(x-2) \cdot pq^x$$

$$= 0 + 0 + 0 + 3(2)pq^3 + 4(3)(2)pq^4 + 5(12)pq^5 + \dots$$

$$= 6pq^3 + 24pq^4 + 60pq^5 + \dots$$

$$= 6pq^3 [1 + 4pq + 10q^2 + \dots]$$

$$= 6pq^3 [1-q]^{-4}$$

$$= 6q^3 \cdot p \cdot p^{-4}$$

$$\text{As: } 1-q = p$$

$$= 6q^3 \cdot p^{-3}$$

$$= \frac{6q^3}{p^3}$$

$$E(X^3) = E[X(X-1)(X-2)] + 3E[X(X-1)] + E(X) \quad \text{--- (A)}$$

$$= \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p}$$

$$E(x^3) = \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} + \dots + n^3 \cdot p^n$$

$$= \frac{q}{p} \left[\frac{6q^2}{p^2} + \frac{6q}{p} + 1 \right]$$

$$= \frac{q}{p} \left[\frac{6q^2 + 6pq + p^2}{p^2} \right]$$

$$= \frac{q}{p^3} [6q^2 + 6pq + p^2]$$

$$= \frac{q}{p^3} [6q(q+p) + p^2]$$

$$= \frac{q}{p^3} [6q(1) + p^2]$$

$$E(x^3) = \frac{q}{p^3} (6q + p^2)$$

4th Moment:

$$E(x^4) = E[x(x-1)(x-2)(x-3)] + 6E[x(x-1)(x-2)] + 7E[x(x-1)] + E(x) \quad \text{--- (A)}$$

$$E[x(x-1)(x-2)(x-3)] = \sum_{x=0}^{\infty} x(x-1)(x-2)(x-3) \cdot p q^x$$

$$= 0 + 0 + 0 + 0 + 24pq^4 + 120pq^5 + 360pq^6 + \dots$$

$$= 24pq^4 + 120p^2q^3 + 360p^3q^2 + \dots$$

$$= 24pq^4 [1 + 5q + 15q^2 + \dots]$$

$$= 24pq^4 (1-q)^{-5}$$

$$\text{As: } 1-q=p$$

$$= 24pq^4 \cdot p^{-5}$$

$$= \frac{24q^4}{p^4}$$

$$E(x^4) = E[x(x-1)(x-2)(x-3)] + 6E[x(x-1)(x-2)] + 7E[x(x-1)] + E(x) \quad \text{--- (A)}$$

$$= \frac{24q^4}{p^4} + \frac{36q^3}{p^3} + \frac{14q^2}{p^2} + \frac{q}{p}$$

Moments About Mean

1st Moment:-

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$= \frac{q}{p} - \frac{q}{p}$$

$$= \frac{q}{p} - \frac{q}{p} = 0$$

2nd Moment:-

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$= \frac{q}{p} \left[\frac{2q}{p} + 1 \right] - \left(\frac{q}{p} \right)^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$

$$= \frac{q}{p} \left[\frac{2q}{p} + 1 - \frac{q}{p} \right]$$

$$= \frac{q}{p} \left[\frac{2q + p - q}{p} \right]$$

$$= \frac{q}{p^2} [2q - q + p]$$

$$= \frac{q}{p^2} [q + p]$$

$$= \frac{q}{p^2} (1)$$

$$\mu_2 = \frac{q}{p^2}$$

3rd Moment:-

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$$

$$= \frac{q}{p^3} (6q + p^2) - \frac{3q}{p} \left[\frac{2q}{p} + 1 \right] \left[\frac{q}{p} \right] + 2 \left(\frac{q}{p} \right)^3$$

$$= \frac{q}{p^3} (6q + p^2) - \left[\frac{6q^2}{p^2} + \frac{3q}{p} \right] \left[\frac{q}{p} \right] + \frac{2q^3}{p^3}$$

$$= \frac{q}{p^3} (6q + p^2) - \frac{6q^3}{p^3} - \frac{3q^2}{p^2} + \frac{2q^3}{p^3}$$

$$= \frac{q}{p^3} \left[\frac{6q + p^2}{p} - \frac{6q^2}{p} - 3q + \frac{2q^2}{p} \right]$$

$$= \frac{q}{p^3} \left[\frac{6q + p^2 - 6q^2 - 3pq + 2q^2}{p} \right]$$

$$= \frac{q}{p^3} \left[\frac{p^2 - 3pq + 6q - 4q^2}{p} \right]$$

$$= \frac{q}{p^3} [p^2 + q^2 - 2pq - pq - 5q^2 + 6q]$$

$$= \frac{q}{p^3} [(q+p)^2 - pq - 5q^2 + 6q]$$

$$= \frac{q}{p^3} [1 - q(p + 5q + 6)]$$

$$= \frac{q}{p^3} [1 - q(p + 5q + 6)]$$

4th Moment:

$$\begin{aligned} \mu_4 &= \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4 \\ &= \frac{24q^4}{p^4} - 4\left[\frac{q}{p^3}(6q+p^2)\right] + 6\left[\frac{2q^2}{p^2} + \frac{q}{p}\right]\left[\frac{q}{p}\right]^2 \\ &\quad - 3\left[\frac{q}{p}\right]^4 \\ &= \frac{24q^4}{p^4} - 4\left[\frac{6q^2+p^2q}{p^3}\right] + 6\left[\frac{2q^2+pq}{p^2}\right]\left[\frac{q^2}{p^2}\right] - 3\left[\frac{q^4}{p^4}\right] \\ &= \frac{24q^4}{p^4} - \frac{24q^2+4p^2q}{p^3} + \frac{12q^4+6pq^3}{p^4} - \frac{q^4}{p^4} \\ &= \frac{24q^4+12q^4+6pq^3-q}{p^4} - \frac{24q^2+4p^2q}{p^3} \\ &= \frac{36q^4+6pq^3-q}{p^4} - \frac{4q}{p^3}\left[6q+p^2\right] \\ &= \frac{q}{p^3}\left[\frac{36q^3+6pq^2-1}{p} - 24q^2-4p^2q\right] \end{aligned}$$

$$\mu_4 = \frac{q}{p^3} \left[\frac{36q^3 + 6pq^2 - 1 - 24pq - 4p^3q}{p} \right]$$

$$= \frac{q}{p^4} (36q^3 + 6pq^2 - 1 - 24pq - 4p^3q)$$

Skewness of Geometric Dist.

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

$$= \frac{q(1 - q(p + 5q + 6))}{p^3}$$

$$= \frac{\left(\frac{q}{p^2}\right)^{3/2} q(1 - q(p + 5q + 6))}{p^3}$$

$$= \frac{1 - q(p + 5q + 6)}{\sqrt{q}}$$

Kurtosis of Geometric Dist.

$$\text{Kurtosis} = \frac{\mu_4}{(\mu_2)^2}$$

$$= \frac{q \cdot (36q^3 - 18pq^2 - 4p^3q - 1)}{p^4}$$

$$= \frac{\left(\frac{q}{p}\right)^2 \cdot q \cdot (36q^3 - 18pq^2 - 4p^3q - 1)}{\frac{q^2}{p^4}}$$

$$\text{Kurtosis} = \frac{(36q^3 - 18pq^2 - 4p^3q - 1)}{q}$$

Probability Generating Function

Geometric Distribution

$$P(X=x) = pq^x$$

P.D.F of Geometric Distribution

$$G(\theta) = E(\theta^x) = \sum_{x=0}^{\infty} pq^x \theta^x$$

$$= p \sum_{x=0}^{\infty} (q\theta)^x$$

$$= p \left[1 + q\theta + \frac{(q\theta)^2}{2!} + \frac{(q\theta)^3}{3!} + \dots \right]$$

$$= p(1 - q\theta)^{-1}$$

$$G(\theta) = \frac{p}{(1 - q\theta)^1}$$

1st Moment:-

$$G'(\theta) = \frac{d}{d\theta} (p(1 - q\theta)^{-1})$$

$$= p(-1)(1 - q\theta)^{-2} (0 - q)$$

$$= pq(1 - q\theta)^{-2}$$

$$\text{put } \theta = 1$$

$$= pq(1 - q)^{-2}$$

$$= pq p^{-2}$$

$$= q p^{-1}$$

$$G'(\theta) = \frac{q}{p}$$

2nd Moment:-

$$G''(\theta) = \frac{d}{d\theta} (pq (1-q\theta)^{-2})$$

$$= pq (-2) (1-q\theta)^{-3} (\theta - q)$$

$$= 2pq^2 (1-q\theta)^{-3}$$

$$\text{put } \theta = 1$$

$$= 2pq^2 (1-q)^{-3}$$

$$= 2pq^2 p^{-3}$$

$$= 2q^2 p^{-2}$$

$$G''(\theta) = \frac{2q^2}{p^2}$$

Variance:

$$V(X) = G'' + G' - (G')^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$

$$= \frac{q}{p} \left[\frac{2q}{p} + 1 - \frac{q}{p} \right]$$

$$= \frac{q}{p} \left[\frac{2q + p - q}{p} \right]$$

$$= \frac{q}{p^2} (2q + p - q)$$

3rd Moment:-

$$G'''(\theta) = \frac{d}{d\theta} (2pq^2(1-q\theta)^{-3})$$

$$= 2pq^2(-3)(1-q\theta)^{-4}(0-q)$$

$$= 6pq^3(1-q\theta)^{-4}$$

$$\text{put } \theta = 1$$

$$= 6pq^3(1-q)^{-4}$$

$$= 6pq^3p^{-4}$$

$$= 6q^3p^{-1}$$

$$G'''(\theta) = \frac{6q^3}{p}$$

4th Moment:-

$$G^{(4)}(\theta) = \frac{d}{d\theta} (6pq^3(1-q\theta)^{-4})$$

$$= 6pq^3(-4)(1-q\theta)^{-5}(0-q)$$

$$= 24pq^4(1-q\theta)^{-5}$$

$$\text{put } \theta = 1$$

$$= 24pq^4(1-q)^{-5}$$

$$= 24pq^4p^{-5}$$

$$= 24q^4p^{-1}$$

$$G^{(4)}(\theta) = \frac{24q^4}{p}$$

Moments Generating Function

Geometric Distribution

$$M_0(t) = (E)^{tn} = \sum_{n=0}^{\infty} p q^n e^{tn}$$

$$= p \sum_{n=0}^{\infty} (q e^t)^n$$

$$= p \left[(q e^t)^0 + \frac{(q e^t)^1}{1!} + \frac{(q e^t)^2}{2!} + \dots \right]$$

$$= p (1 + q e^t + \dots)$$

$$M_0(t) = p (1 - q e^t)^{-1}$$

1st. Moment:-

$$M'_0(t) = p (-1) (1 - q e^t)^{-2} (-q e^t)$$
$$= p q e^t (1 - q e^t)^{-2}$$

$$\text{put } t=0$$

$$= p q (1 - q)^{-2}$$

$$= pq(p^{-2})$$

$$E(x) = \frac{q}{p}$$

2nd Moment:-

$$pqe^t(1-qe^t)^{-1}$$

$$M_0''(t) = pqe^t(1-qe^t)^{-2} + pqe^t(-2)(1-qe^t)^{-3}(-qe^t)$$

$$= pqe^t(1-qe^t)^{-2} + 2pq^2e^{2t}(1-qe^t)^{-3}$$

$$\text{put } t=0$$

$$= pqe^0(1-q)^{-2} + 2pq^2e^0(1-qe^0)^{-3}$$

$$= pq(p^{-2}) + 2pq^2(p^{-3})$$

$$E(x^2) = \frac{q}{p} + 2\frac{q^2}{p^2}$$

3rd Moment:-

$$pqe^t(1-qe^t)^{-2} + 2pq^2e^{2t}(1-qe^t)^{-3}$$

$$M_0'''(t) = pqe^t(1-qe^t)^{-2} + pqe^t(-2)(1-qe^t)^{-3}(-qe^t) + 2$$

$$pq^2e^{2t}(1-qe^t)^{-3} + 2pq^2e^{2t}(-3)(1-qe^t)^{-4}(-qe^t)$$

$$= pqe^t(1-qe^t)^{-2} + 2pq^2e^{2t}(1-qe^t)^{-3} + 4pq^3e^{3t}(1-qe^t)^{-4}$$

$$+ 6pq^3e^{3t}(1-qe^t)^{-4}$$

$$\text{put } t=0$$

$$= p q (1-q)^{-2} + 2 p q^2 (1-q)^{-3} + 4 p q^2 (1-q)^{-3} + 6 p q^3 (1-q)^{-4}$$

$$E(X^3) = \frac{q}{p} + 2 \frac{q^2}{p^2} + 4 \frac{q^2}{p^2} + 6 \frac{q^3}{p^3}$$

4th Moment:-

$$p q e^t (1-q e^t)^{-2} + 2 p q^2 e^{2t} (1-q e^t)^{-3} + 4 p q^2 e^{2t} (1-q e^t)^{-3} + 6 p q^3 e^{3t} (1-q e^t)^{-4}$$

$$\begin{aligned} M_0''(t) &= p q e^t (1-q e^t)^{-2} + p q e^t (-2) (1-q e^t)^{-3} (-q e^t) + \\ & 2 p q^2 (2) (e^{2t}) (1-q e^t)^{-3} + 2 p q^2 e^{2t} (-3) (1-q e^t)^{-4} \\ & (-q e^t) + 4 p q^2 (2) e^{2t} (1-q e^t)^{-3} + 4 p q^2 e^{2t} (-3) \\ & (1-q e^t)^{-4} (-q e^t) + 6 p q^3 (3) e^{3t} (1-q e^t)^{-4} + \\ & 6 p q^3 e^{3t} (-4) (1-q e^t)^{-5} (-q e^t) \\ &= p q e^t (1-q e^t)^{-2} + 2 p q^2 e^{2t} (1-q e^t)^{-3} + 4 p q^2 e^{2t} (1-q e^t)^{-3} \\ & 6 p q^3 e^{3t} (1-q e^t)^{-4} + 8 p q^2 e^{2t} (1-q e^t)^{-3} + 12 p q^3 e^{3t} \\ & (1-q e^t)^{-4} + 18 p q^3 e^{3t} (1-q e^t)^{-4} + 24 p q^4 e^{4t} (1-q e^t)^{-5} \end{aligned}$$

put $t=0$

$$\begin{aligned} &= p q (1-q)^{-2} + 2 p q^2 (1-q)^{-3} + 4 p q^2 (1-q)^{-3} + 6 \\ & p q^3 (1-q)^{-4} + 8 p q^2 (1-q)^{-3} + 12 p q^3 (1-q)^{-4} + \\ & 18 p q^3 (1-q)^{-4} + 24 p q^4 (1-q)^{-5} \end{aligned}$$

$$\begin{aligned}
 &= p q (p^{-2}) + 2 p q^2 (p^{-3}) + 4 p q^2 (p^{-3}) + 6 p q^3 (p^{-4}) \\
 &\quad + 8 p q^3 (p^{-4}) + 12 p q^3 (p^{-4}) + 18 p q^3 (p^{-4}) + 24 p q^4 (p^{-5}) \\
 &= p q / p + 2 q^2 / p^2 + 4 q^2 / p^2 + 6 q^3 / p^3 + 8 q^3 / p^3 + 12 q^3 / p^3 + 18 q^3 / p^3 + 24 q^4 / p^4
 \end{aligned}$$

$$E(x^4) = \frac{q}{p} + 14 \frac{q^2}{p^2} + 36 \frac{q^3}{p^3} + 24 \frac{q^4}{p^4}$$

Moment About Mean

1st Moment:-

$$\begin{aligned}
 \mu_1 &= \mu'_1 - \mu'_1 = 0 \\
 &= \frac{q}{p} - \frac{q}{p} = 0
 \end{aligned}$$

2nd Moment:-

$$\begin{aligned}
 \mu_2 &= \mu'_2 - (\mu'_1)^2 \\
 &= \frac{q}{p} + 2 \frac{q^2}{p^2} - \frac{q^2}{p^2}
 \end{aligned}$$

$$\mu_2 = \frac{q}{p} + \frac{q^2}{p^2}$$

$$\mu_2 = \frac{q}{p} \left(1 + \frac{q}{p} \right)$$

3rd Moment:

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$= \left(\frac{q}{p} + \frac{6q^2}{p^2} + \frac{6q^3}{p^3} \right) - 3 \left(\frac{q}{p} + \frac{2q^2}{p^2} \right) + \left(\frac{2q^3}{p^3} \right)$$

$$= \left(\frac{q}{p} + \cancel{\frac{6q^2}{p^2}} + \frac{6q^3}{p^3} - 3 \frac{q}{p} - \cancel{\frac{6q^2}{p^2}} + 2 \frac{q^3}{p^3} \right)$$

$$= \left(\frac{6q^3}{p^3} - \frac{2q}{p} \right) = \left(\frac{6q^3 - 2qp^2}{p^3} \right)$$

$$= \frac{2q}{p^3} (3q^2 - p^2)$$

$$E(Y^3) = \left(\frac{2q}{p^3} (3q^2 - p^2) \right)$$

4th Moment

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

$$= \frac{q}{p} + \frac{14q^2}{p^2} + \frac{36q^3}{p^3} + \frac{24q^4}{p^4} - 4\left(\frac{q}{p} + \frac{6q^2}{p^2} + \frac{6q^3}{p^3}\right)$$

$$+ 6\left(\left(\frac{q}{p} + \frac{2q^2}{p^2}\right)\left(\frac{q^2}{p^2}\right)\right) - 3\frac{q^4}{p^4}$$

$$= \frac{q}{p} + \frac{14q^2}{p^2} + \frac{36q^3}{p^3} + \frac{24q^4}{p^4} - 4\frac{q}{p} - \frac{24q^2}{p^2} - \frac{24q^3}{p^3}$$

$$+ 6\frac{q^3}{p^3} + 12\frac{q^4}{p^4} - 3\frac{q^4}{p^4}$$

$$= \frac{33q^4}{p^4} + \frac{18q^3}{p^3} - \frac{10q^2}{p^2} - \frac{3q}{p}$$

$$= \frac{q}{p} \left[\frac{33q^3}{p^3} + \frac{18q^2}{p^2} - \frac{10q}{p} - 3 \right]$$

$$= \frac{q}{p} \left[\frac{33q^3 + 18pq^2 - 10p^2q - 3p^3}{p^3} \right]$$

$$= \frac{q}{p^4} \left[33q^3 + 18pq^2 - 10p^2q - 3p^3 \right]$$

$$= \frac{q}{p^4} \left[q^2(33q + 18p) - p^2(10q + 3p) \right]$$

$$= \frac{q}{p^4} \left[q^2 (32q + 17p + q + p) - p^2 (9q + 2p + q + p) \right]$$

$$= \frac{q}{p^4} \left[q^2 (32q + 17p) - p^2 (9q + 2p) \right]$$

$$= \frac{q}{p^4} \left[q^2 (32q + 16p + p) - p^2 (8q + p + q + p) \right]$$

$$= \frac{q}{p^4} \left[q^2 [16(2q + p) + p] - p^2 (8q + p) \right]$$

$$= \frac{q}{p^4} \left[q^2 [16(q + q + p) + p] - p^2 (7q + q + p) \right]$$

$$= \frac{q}{p^4} \left[q^2 [16(q) + p] - p^2 (7q) \right]$$

$$= \frac{q}{p^4} \left[q^2 [15q + q + p] - 7p^2 q \right]$$

$$= \frac{q}{p^4} \left[15q^3 - 7p^2 q \right]$$

$$= \frac{q}{p^4} \left[q (15q^2 - 7p^2) \right]$$

$$E(x^4) = \frac{q}{p^4} \left[q (15q^2 - 7p^2) \right]$$

Geometric Distribution.

Factorial G.F of Geometric Distribution

$$H_0(x) = p(p-qx)^{-1}$$

By taking log of F.G.F we get

$$H_0(x) = \log p + \log(p-qx)^{-1}$$

$$\text{Taking derivative w.r.t } x$$

By taking derivative w.r.t (x)

1st:-

PUACP

$$h'_0(x) = 0 - (p-qx)^{-1}(-q)$$

$$= q(p-qx)^{-1}$$

$$\text{put } x = 0$$

$$= q(p)^{-1}$$

$$K_0(1) = q/p$$

$$q(p - q\alpha)^{-1}$$

$$\begin{aligned} h_0''(\alpha) &= q(-1)(p - q\alpha)^{-2}(-q) \\ &= q^2(p - q\alpha)^{-2} \end{aligned}$$

$$\text{put } \alpha = 0$$

$$= q^2(p)^{-2}$$

$$= q^2/p^2$$

3rd:-

$$q^2(p - q\alpha)^{-2}$$

$$\begin{aligned} h_0'''(\alpha) &= q^2(-2)(p - q\alpha)^{-3}(-q) \\ &= +2q^3(p - q\alpha)^{-3} \end{aligned}$$

$$\text{put } \alpha = 0$$

$$= +2q^3(p)^{-3}$$

$$= 2q^3/p^3$$

4th:-

$$2q^3(p - q\alpha)^{-3}$$

$$\begin{aligned} h_0^{(4)}(\alpha) &= 2q^3(-3)(p - q\alpha)^{-4}(-q) \\ &= 6q^4(p - q\alpha)^{-4} \end{aligned}$$

$$\text{put } \alpha = 0$$

$$= 6q^4(p)^{-4}$$

$$= 6q^4/p^4$$

Cumulatives (Kapas)

1st:-

$$K_1 = K_{e(1)} = m_1 \\ = q/p$$

2nd:-

$$K_2 = K_{e(1)} + K_{e(2)} = u_1$$

$$= \frac{q}{p} + \frac{q^2}{p^2}$$

$$= \frac{qp + q^2}{p^2}$$

$$= \frac{q(p+q)}{p^2}$$

$$\because p+q=1$$

$$= \frac{q}{p^2} = u_1$$

3rd:-

$$K_3 = K_{e(1)} + 3K_{e(2)} + K_{e(3)} = u_2$$

$$= \frac{q}{p} + 3\frac{q^2}{p^2} + \frac{2q^3}{p^3}$$

$$= \frac{p^2q + 3pq^2 + 2q^3}{p^3}$$

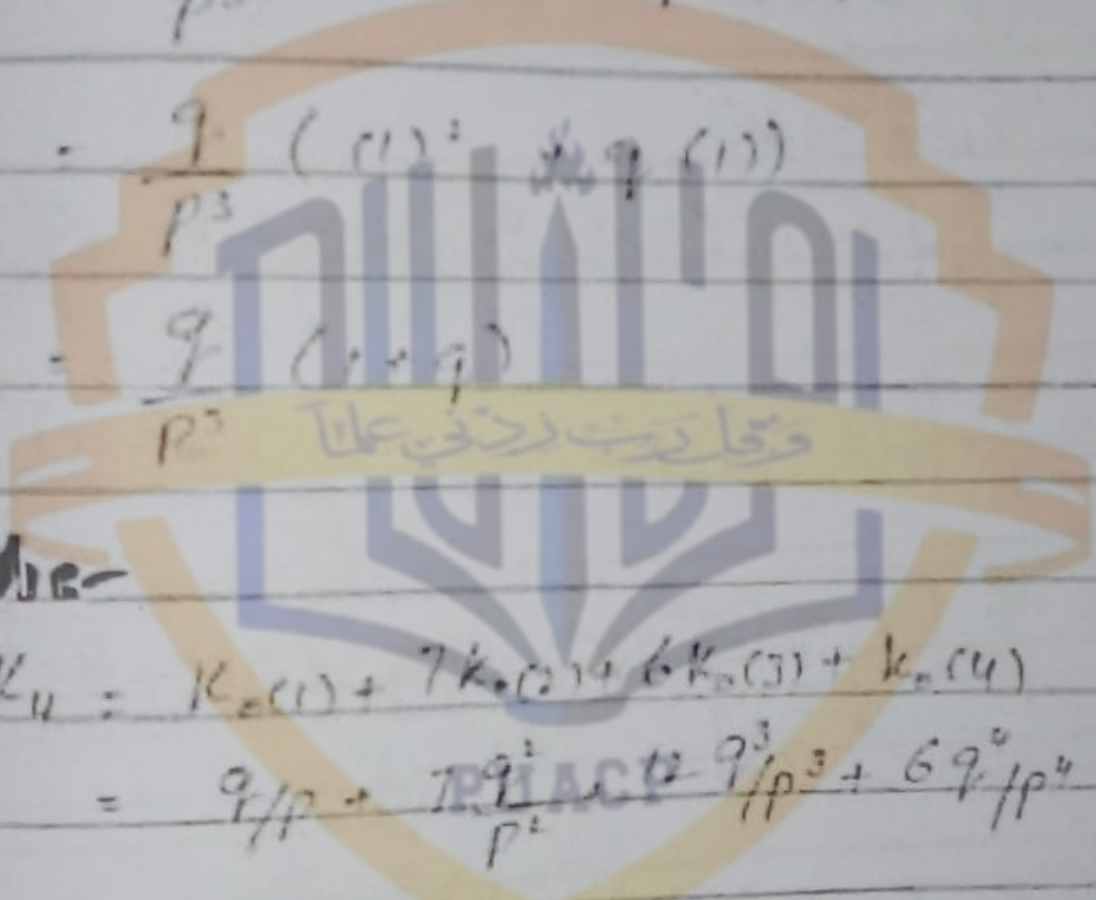
$$= \frac{q}{p^3} (p^3 + 3pq + 2q^2)$$

$$= \frac{q}{p^3} (p^3 + 2pq + q^2 + pq + q^2)$$

$$= \frac{q}{p^3} ((q+p)^2 + q(q+p))$$

$$= \frac{q}{p^3} (1)^2 + q(1)$$

$$u_3 = \frac{q}{p^3} (1+q)$$

4th-


$$K_u = K_e(1) + 7K_e(2) + 6K_e(3) + K_e(4)$$

$$= \frac{q}{p} + \frac{7q^2}{p^2} + \frac{12q^3}{p^3} + \frac{6q^4}{p^4}$$

$$= \frac{p^3q + 7p^2q^2 + 12pq^3 + 6q^4}{p^4}$$

$$= \frac{q}{p^4} (p^3 + 7p^2q + 12pq^2 + 6q^3)$$

$$= \frac{q}{p^4} [q^3 + 3pq^2 + 3p^2q + p^3 + 4p^2q + 8pq^2 + 5q^3]$$

$$= \frac{q}{p^4} [(q+p)^3 + 4p^2q + 8pq^2 + 5q^3]$$

$$= \frac{q}{p^4} [1 + q(4p^2 + 8pq + 5q^2)]$$

$$u_4 = k_4 + 3k_1$$

$$= \frac{q}{p^4} [1 + q(4p^2 + 8pq + 5q^2)] + 3 \frac{q^4}{p^4}$$

$$= \frac{q}{p^4} [1 + q(4p^2 + 8pq + 5q^2) + 3q^3]$$

$$= \frac{q}{p^4} [1 + q(2q^2 + p^2 + 8pq + 3p^2 + 3q^2 + 1pq) + 3q]$$

$$= \frac{q}{p^4} [1 + q((q+p)^2 + 2p^2 + 2q^2 + q^2 + pq) + 3q]$$

$$= \frac{q}{p^4} [1 + q(4(1)^2 + 2p^2 + 2q^2 + q(p+q) + 3q)]$$

$$= \frac{q}{p^4} [1 + q(4 + 2p^2 + 2q^2 + q + 3q)]$$

$$= \frac{q}{p^4} [1 + q(4 + 2p^2 + q(2q+1+3))]$$

$$= \frac{q}{p^4} [1 + q(4 + 2p^2 + q(2q + 4))]$$

$$= \frac{q}{p^4} [1 + q(4 + 2p^2 + 2q(q + 2))]$$

$$\mu_4 = \frac{q}{p^4} [1 + 2q(2 + p^2 + q(q + 2))]$$

Skewness

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

$$= \frac{q/p^3(1+q)}{(q/p^2)^{3/2}}$$

$$= \frac{q/p^3(1+q)}{q/p^2 \cdot \frac{\sqrt{q}}{p}}$$

$$= \frac{(1+q)/p}{\sqrt{q}/p}$$

$$\text{Skewness} = \frac{1+q}{\sqrt{q}}$$

Hyperbolicity:

$$\begin{aligned} \text{Hess} f &= \frac{1}{(11)^2} \\ &= \frac{1}{p^4} [1 + 2q(2 + p^2 + q(q+2))] \\ &= \frac{1 + 2q(2 + p^2 + q(q+2))}{p^4} \end{aligned}$$

Factorial Generating Function Geometric Distribution

$$M_0(t) = p(1 - qe^t)^{-1}$$

replace e^t by $1 + \alpha$

$$H_0(t) = p(1 - q(1 + \alpha))^{-1}$$

$$= p(1 - q - q\alpha)^{-1}$$

$$H_0(t) = p(p - q\alpha)^{-1}$$

By taking derivatives to get
expected values.

$$H_0 = p(p - q\alpha)^{-1}$$

1st: $H'_0 = p(-1)(p - q\alpha)^{-2}(-q)$

$$= pq(p - q\alpha)^{-2}$$

put $\alpha = 0$

$$= pq(p)^{-2}$$

$$= \frac{q}{p}$$

$$E(x) = q/p$$

2nd: $pq(p-q\alpha)^{-2}$
 $H' = (pq)(-2)(p-q\alpha)^{-3}(-q)$
 $= 2pq^2(p-q\alpha)^{-3}$

put $\alpha = 0$

$= 2pq^2(p)^{-3}$

$= \frac{2q^2}{p^2}$

$E(x(x-1)) = \frac{2q^2}{p^2}$

3rd: $2pq^2(p-q\alpha)^{-3}$

$H'' = 2pq^2(-3)(p-q\alpha)^{-4}(-q)$

$= 6pq^3(p-q\alpha)^{-4}$

put $\alpha = 0$

$= 6pq^3(p)^{-4}$

$= \frac{6q^3}{p^3}$

$E(x(x-1)(x-2)) = \frac{6q^3}{p^3}$

4th: $6pq^3(p-q\alpha)^{-4}$

$H''' = 6pq^3(-4)(p-q\alpha)^{-5}(-q)$

$= 24pq^4(p-q\alpha)^{-5}$

put $\alpha = 0$

$$= 24pq^4 (p)^{-5}$$

$$= \frac{24q^4}{p^4}$$

$$E(x(x-1)(x-2)(x-3)) = \frac{24q^4}{p^4}$$

Now we find the ordinary moments

$$\mu'_1 = E(x) = q/p$$

$$\begin{aligned} \mu'_2 = E(x^2) &= E(x(x-1)) + E(x) \\ &= \frac{2q^2}{p^2} + \frac{q}{p} \end{aligned}$$

$$\begin{aligned} \mu'_3 = E(x^3) &= E(x(x-1)(x-2)) + 3E(x(x-1)) + E(x) \\ &= \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} \end{aligned}$$

$$\begin{aligned} \mu'_4 = E(x^4) &= E(x(x-1)(x-2)(x-3)) + 6E(x(x-1)(x-2)) \\ &\quad + 7E(x(x-1)) + E(x) \\ &= \frac{24q^4}{p^4} + \frac{36q^3}{p^3} + \frac{14q^2}{p^2} + \frac{q}{p} \end{aligned}$$

Moments About Mean

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$\frac{q}{p} - \frac{q}{p} = 0$$

$$\frac{q}{p} - \frac{q}{p} = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$

$$= \frac{2q^2 + pq - q^2}{p^2} = \frac{pq + q^2}{p^2}$$

$$= \frac{q(p+q)}{p^2} = \frac{q}{p^2} \quad \because p+q=1$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$= \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} - 3\left(\frac{2q^2}{p^2} + \frac{q}{p}\right)\left(\frac{q}{p}\right) + 2\frac{q^3}{p^3}$$

$$= \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} - 3\left[\frac{2q^3}{p^3} + \frac{q^2}{p^2}\right] + \frac{2q^3}{p^3}$$

$$= \frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} - \frac{6q^3}{p^3} - \frac{3q^2}{p^2} + \frac{2q^3}{p^3}$$

$$u_3 = \frac{2q^3}{p^3} + \frac{3q^2}{p^2} + \frac{q}{p}$$

$$= \frac{q}{p} \left[\frac{2q^2}{p^2} + \frac{3q}{p} + 1 \right]$$

$$= \frac{q}{p} \left[\frac{2q^2 + 3pq + p^2}{p^2} \right]$$

$$= \frac{q}{p^2} [2q^2 + 3pq + p^2]$$

$$= \frac{q}{p^2} [q^2 + 2qp + p^2 + q^2 + qp]$$

$$= \frac{q}{p^2} [(q+p)^2 + q(q+p)] \quad \because q+p=1$$

$$= \frac{q}{p^2} [1 + q]$$

$$u_4 = \frac{24q^4}{p^4} + \frac{36q^3}{p^3} + \frac{14q^2}{p^2} + \frac{q}{p} - 4 \left[\left(\frac{q}{p} \right) \left(\frac{6q^3}{p^3} + \frac{6q^2}{p^2} + \frac{q}{p} \right) \right]$$

$$+ 6 \left[\frac{q^2}{p^2} \left(\frac{2q^2}{p^2} + \frac{q}{p} \right) \right] - 3 \frac{q^4}{p^4}$$

$$= \frac{24q^4}{p^4} + \frac{36q^3}{p^3} + \frac{14q^2}{p^2} + \frac{q}{p} - \frac{24q^4}{p^4} - \frac{24q^3}{p^3} - \frac{4q^2}{p^2}$$

$$+ \frac{12q^4}{p^4} + \frac{6q^3}{p^3} - 3 \frac{q^4}{p^4}$$

$$= \frac{12q^4}{p^4} - \frac{3q^3}{p^4} + \frac{18pq^3}{p^3} + \frac{10q^2}{p^2} + \frac{q}{p}$$

$$= \frac{9q^4}{p^4} + \frac{18q^3}{p^3} + \frac{10q^2}{p^2} + \frac{q}{p}$$

$$= \frac{q}{p} \left[\frac{9q^3}{p^3} + \frac{18q^2}{p^2} + \frac{10q}{p} + 1 \right]$$

$$= \frac{q}{p} \left[\frac{9q^3 + 18pq^2 + 10p^2q + p^3}{p^3} \right]$$

$$= \frac{q}{p^4} [9q^3 + 18pq^2 + 10p^2q + p^3]$$

$$= \frac{q}{p^4} [q^2(q + 2p) + p(10q + p)]$$

$$= \frac{q}{p^4} [q^3 + 3pq^2 + 3p^2q + p^3 + 8q^3 + 15pq^2 + 7p^2q]$$

$$= \frac{q}{p^4} [(q+p)^3 + q(8q^2 + 15pq + 7p^2)]$$

$$= \frac{q}{p^4} [1 + q(8q^2 + 15pq + 7p^2)]$$

Cumulents Generating Function

Geometric Distribution

Moment Generating Function of Geometric Distribution :

$$M_0(t) = p(1 - qe^t)^{-1}$$

By taking log of M.G.F we get :

$$\log M_0(t) = \log(p(1 - qe^t)^{-1})$$

$$C_0(t) = p(1 - qe^t)^{-1}$$

1st:-

$$C'_0(t) = p(-1)(1 - qe^t)^{-2}(-qe^t)$$

$$= pqe^t(1 - qe^t)^{-2}$$

$$\text{put } t=0$$

$$= pq(1 - q)^{-2}$$

$$= pq(p^{-2})$$

$$\mu_1 = \frac{q}{p} = \mu_1$$

$$\therefore p q e^t (1 - q e^t)^2 \therefore \dots$$

2nd:

$$C_0'(t) = p q e^t (1 - q e^t)^{-2} + p q e^t (-2) (1 - q e^t)^{-3} (-q e^t) \\ = p q e^t (1 - q e^t)^{-2} + 2 p q^2 e^{2t} (1 - q e^t)^{-3}$$

$$\text{put } t = 0$$

$$= p q (1 - q)^{-2} + 2 p q^2 (1 - q)^{-3}$$

$$= p q (p^{-2}) + 2 p q^2 (p^{-3})$$

$$\mu_2 = \frac{q}{p} + 2 \frac{q^2}{p^2} = \mu_2$$

3rd:

$$p q e^t (1 - q e^t)^2 + 2 p q^2 e^{2t} (1 - q e^t)^{-3}$$

$$C_0'''(t) = p q e^t (1 - q e^t)^{-2} + p q e^t (-2) (1 - q e^t)^{-3} (-q e^t) + \\ 2 p q^2 2 e^{2t} (1 - q e^t)^{-3} + 2 p q^2 e^{2t} (-3) (1 - q e^t)^{-4} (-q e^t) \\ = p q e^t (1 - q e^t)^{-2} + 2 p q^2 e^{2t} (1 - q e^t)^{-3} + 4 p q^2 e^{2t} \\ (1 - q e^t)^{-3} + 6 p q^3 e^{3t} (1 - q e^t)^{-4}$$

$$\text{put } t = 0$$

$$= p q (1 - q)^{-2} + 2 p q^2 (1 - q)^{-3} + 4 p q^2 (1 - q)^{-3} + 6 p q^3 (1 - q)^{-4}$$

$$= p q (p^{-2}) + 2 p q^2 (p^{-3}) + 4 p q^2 (p^{-3}) + 6 p q^3 (p^{-4})$$

$$= \frac{q}{p} + 2 \frac{q^2}{p^2} + 4 \frac{q^2}{p^2} + 6 \frac{q^3}{p^3}$$

$$u = \frac{q}{r} + 6\frac{q^2}{r^2} + 6\frac{q^3}{r^3} + \dots$$

4th:

$$0 \cdot q q^2 (1 - q e^t)^{-2} + 2 p q^2 e^{2t} (1 - q e^t)^{-3} + 4 p q^2 e^{2t} (1 - q e^t)^{-3} + 6 p q^3 e^{3t} (1 - q e^t)^{-4}$$

$$\begin{aligned} & 0 \cdot p q q^2 (1 - q e^t)^{-2} + p q e^t (-2) (1 - q e^t)^{-3} (-q e^t) + 2 p q^2 e^{2t} (1 - q e^t)^{-3} \\ & + 2 p q^2 e^{2t} (-3) (1 - q e^t)^{-4} (-q e^t) + 4 p q^2 e^{2t} (1 - q e^t)^{-3} \\ & + 4 p q^2 e^{2t} (-3) (1 - q e^t)^{-4} (-q e^t) + 6 p q^3 e^{3t} (1 - q e^t)^{-4} \\ & + 6 p q^3 e^{3t} (-4) (1 - q e^t)^{-5} (-q e^t) \end{aligned}$$

$$\begin{aligned} & = p q e^t (1 - q e^t)^{-2} + 2 p q^2 e^{2t} (1 - q e^t)^{-3} + 4 p q^2 e^{2t} (1 - q e^t)^{-3} \\ & + 6 p q^3 e^{3t} (1 - q e^t)^{-4} + 8 p q^2 e^{2t} (1 - q e^t)^{-3} + 12 p q^3 e^{3t} (1 - q e^t)^{-4} \\ & + 8 p q^3 e^{3t} (1 - q e^t)^{-4} + 24 p q^4 e^{4t} (1 - q e^t)^{-5} \end{aligned}$$

put $t = 0$

$$\begin{aligned} & = p q (1 - q)^{-2} + 2 p q^2 (1 - q)^{-3} + 4 p q^2 (1 - q)^{-3} + 6 p q^3 (1 - q)^{-4} \\ & + 8 p q^2 (1 - q)^{-3} + 12 p q^3 (1 - q)^{-4} + 8 p q^3 (1 - q)^{-4} + 24 p q^4 (1 - q)^{-5} \end{aligned}$$

$$\begin{aligned} & = p q (p^{-2}) + 2 p q^2 (p^{-3}) + 4 p q^2 (p^{-3}) + 6 p q^3 (p^{-4}) \\ & + 8 p q^2 (p^{-3}) + 12 p q^3 (p^{-4}) + 8 p q^3 (p^{-4}) + 24 p q^4 (p^{-5}) \end{aligned}$$

$$\begin{aligned} & = \frac{q}{p} + 2 \frac{q^2}{p^2} + 4 \frac{q^2}{p^2} + 6 \frac{q^3}{p^3} + 8 \frac{q^2}{p^2} + 12 \frac{q^3}{p^3} + 8 \frac{q^3}{p^3} + 24 \frac{q^4}{p^4} \end{aligned}$$

$$K_4 = \frac{q}{p} + 14 \frac{q^2}{p^2} + 36 \frac{q^3}{p^3} + 24 \frac{q^4}{p^4}$$

$$K_4 = K_4 + 3K_2$$

$$= \frac{q}{p} + 14 \frac{q^2}{p^2} + 36 \frac{q^3}{p^3} + 24 \frac{q^4}{p^4} + 3 \frac{q^2}{p^2} + 6 \frac{q^4}{p^4}$$

$$= \frac{q}{p} + 17 \frac{q^2}{p^2} + 36 \frac{q^3}{p^3} + 30 \frac{q^4}{p^4}$$

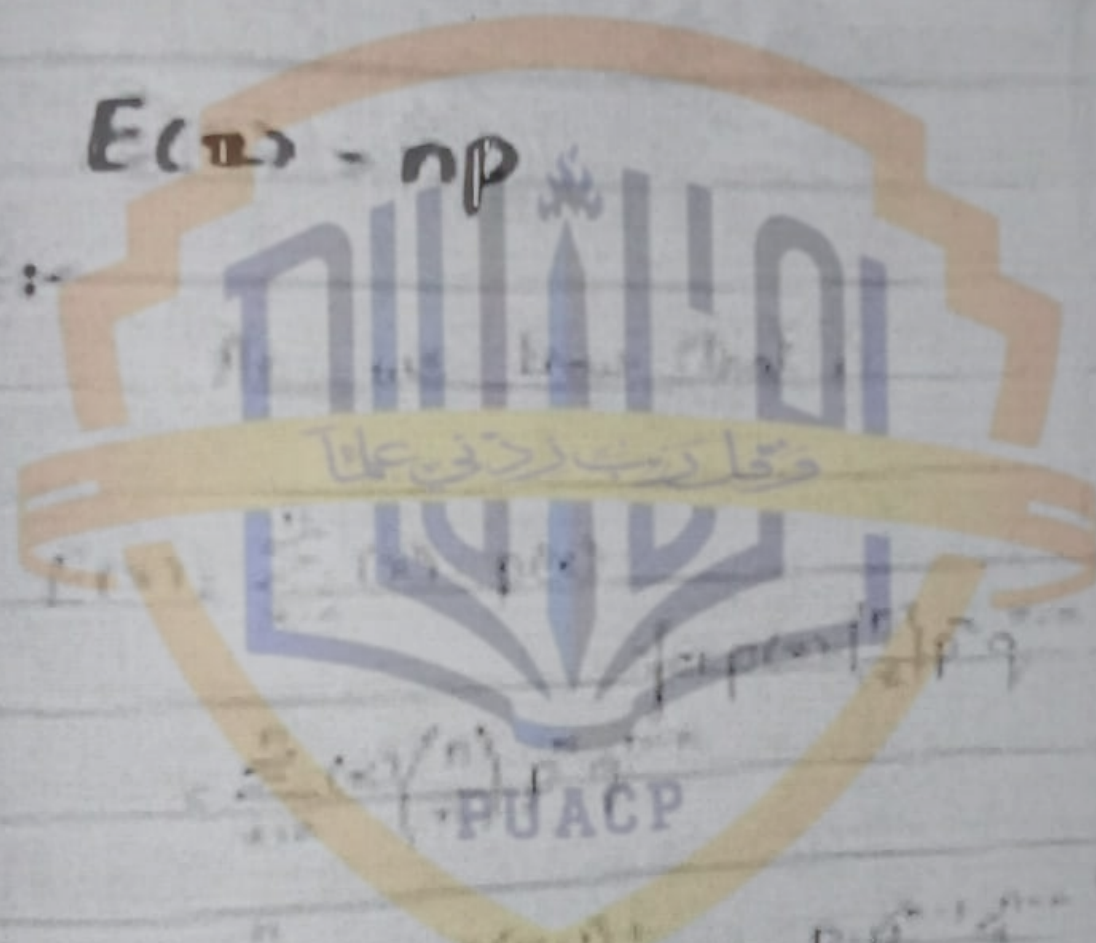
Binomial Distribution

Moments About Origin

1st Moment :-

$$E(x) = np$$

Proof :-



$$E(x) = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=1}^n x \binom{n}{x} p^x q^{n-x}$$

$$= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} q^{n-x}$$

$$= np \left[\binom{n-1}{0} p^0 q^{n-1} + \binom{n-1}{1} p^1 q^{n-2} + \dots + \right.$$

$$\left. \binom{n-1}{n-1} p^{n-1} q^{n-n} \right]$$

As $p+q=1$

$$= np [q^{n-1} + (n-1)pq^{n-2} + \dots + p^{n-1}]$$

$$= np [q+p]^{n-1} \quad \text{By Binomial series}$$

$$= np [1]^{n-1} \quad \text{As } q+p=1$$

$$f'(x) = np$$

So,

$$\mu'_1 = np$$

Hence : proved.

2nd Moment:

$$\mu'_2 = \text{Var}(x) = npq$$

Proof:-

As we know that:

$$\text{Var}(x) = E(x^2) - [E(x)]^2 \quad \text{--- (A)}$$

At first we will solve:-

$$E(x^2) = \sum_{x=0}^n x^2 \cdot p(x)$$

$$= \sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x} \quad \text{--- (1)}$$

By adding and subtracting
x from x^2 we get:

$$x^2 + x - x$$

$$x^2 + x + x$$

$$x(x+1) - x$$

By taking expectation on both sides

$$E(x) = E(x(x-1)) + E(x) \quad \text{--- (1)}$$

By putting $x^2 = x(x-1)$ in equ (1)

$$\begin{aligned} E(x(x-1)) &= \sum_{x=0}^n x(x-1) \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=0}^n x(x-1) \frac{n(n-1)(n-2)!}{x(x-1)(x-2)!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=2}^n \frac{n(n-1)(n-2)!}{(x-2)!(n-x)!} p^2 p^{x-2} q^{n-x} \\ &= n(n-1)p^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} q^{n-x} \\ &= n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x} \\ &= n(n-1)p^2 \left[\binom{n-2}{0} p^0 q^{n-2} + \binom{n-2}{1} p^1 q^{n-3} \right. \\ &\quad \left. + \dots + \binom{n-2}{n-2} p^{n-2} q^{n-n} \right] \end{aligned}$$

$$= n(n-1)p^2 [q+p]^{n-2} \quad \text{By using Binomial series}$$

$$= n(n-1)p^2 [1]^{n-2} \quad \text{As } q+p=1$$

$$E(x(x-1)) = n(n-1)p^2$$

By putting this value in equation (B) we get

$$E(x)^2 = E(x(x-1)) + E(x) \quad \text{--- (B)}$$

$$E(x)^2 = n(n-1)p^2 + np \quad \text{As } E(x) = np$$

$$= (n^2 - n)p^2 + np$$

$$E(x)^2 = n^2p^2 - np^2 + np$$

By putting the value of $E(x)^2$ in equation (A) we get,

$$\begin{aligned} \text{Var}(x) &= E(x)^2 - (E(x))^2 \\ &= n^2p^2 - np^2 + np - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= n^2p^2 - n^2p^2 - np^2 + np \\ &= np - np^2 \end{aligned}$$

$$\text{Var}(x) = np - np^2$$

$$= np(1-p)$$

$$\text{Var}(x) = npq$$

$$n(1-p)q$$

$$\mu'_2 = \text{Var}(x) = npq$$

Hence proved

3rd Moment:

$$\mu'_3 = E(x^3) = n(1-p)$$

As we know that

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$$\mu'_3 = E(x^3) = \sum_{x=0}^n x^3 p(x)$$

So, Now

$$E(x)^3 = \sum_{x=0}^n x^3 \cdot p(x)$$

$$= \sum_{x=0}^n x^3 \binom{n}{x} p^x q^{n-x}$$

$$\text{Here } x^3 = x(x-1)(x-2) + 3x(x-1) + x$$

$$E(x)^3 = \sum_{x=0}^n [x(x-1)(x-2) + 3x(x-1) + x] \cdot p(x)$$

$$U_3' = \left[\sum_{x=0}^n x(x-1)(x-2) + \sum_{x=0}^n (3x(x-1)) + \sum_{x=0}^n x \right] \cdot p(x)$$

$$U_3' = \left[\sum_{x=0}^n x(x-1)(x-2) \cdot p(x) + 3 \sum_{x=0}^n x(x-1) \cdot p(x) + \sum_{x=0}^n x \cdot p(x) \right] \quad \text{--- (1)}$$

Now:

$$\sum_{x=0}^n x(x-1)(x-2) \cdot p(x) \quad \text{--- (2) we solve it.}$$

$$= \sum_{x=0}^n x(x-1)(x-2) \cdot p^x q^{n-x}$$

$$= \sum_{x=0}^n x(x-1)(x-2) \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=3}^n x(x-1)(x-2) \frac{n(n-1)(n-2)(n-3)!}{x!(n-x)!} p^3$$

$$p^{x-3} q^{n-x}$$

$$= \sum_{x=3}^n \frac{n(n-1)(n-2)(n-3)!}{(n-3)!(n-x)!} p^3 \cdot p^{x-3} q^{n-x}$$

$$= n(n-1)(n-2) p^3 \sum_{x=3}^n \frac{(n-3)!}{(n-3)!(n-x)!} p^{x-3} q^{n-x}$$

(3)

$$= n(n-1)(n-2)p^3 \sum_{r=3}^n \binom{n-1}{r-1} p^{n-r} q^{r-n}$$

$$= n(n-1)(n-2)p^3 \left[\binom{n-1}{0} p^3 q^{n-3} + \binom{n-1}{1} p^2 q^{n-4} + \binom{n-1}{2} p^1 q^{n-5} + \dots + \binom{n-1}{n-3} p^4 q^{n-n} \right]$$

$$= n(n-1)(n-2)p^3 \left[q^{n-3} + (n-1)pq^{n-4} + \binom{n-1}{2} p^2 q^{n-5} + \dots + p^{n-3} \right]$$

$$= n(n-1)(n-2)p^3 [q+p]^{n-3} \quad \text{By Binomial Series.}$$

$$= n(n-1)(n-2)p^3 (1)^{n-3} \quad \text{As } q+p=1$$

$$= n(n-1)(n-2)p^3$$

From equation (1) we get.

$$E(x^3) = \sum_{x=0}^n x(x-1)(x+2) + 3E(x)^2 + E(x)$$

By putting values we get.

$$E(x^3) = \sum_{i=1}^n x_i^3 \cdot p_i = \sum_{i=1}^n x_i^3 \cdot p$$

$$E(x^3) = E(x^3)$$

Note: provided.

4th Moment

$$\mu'_4 = E(x^4)$$

As we know that

$$E(x^n) = \sum_{i=1}^n x_i^n \cdot p(x_i)$$

$$= \sum_{i=1}^n x_i^n \cdot p \cdot q^{n-i}$$

$$= \sum_{i=1}^n \left[x(x-1)(x-2)(x-3) + 6x(x-1)(x-2) + x(x-1) \right] \binom{n}{i} p^i q^{n-i} \quad \text{--- (A)}$$

From equation (A) first we calculate

$$\sum_{i=1}^n \left[x(x-1)(x-2)(x-3) \right] \binom{n}{i} p^i q^{n-i}$$

$$= \sum_{x=0}^n x(x-1)(x-2) \frac{1}{x!} p^x q^{n-x}$$

$$= \sum_{x=0}^n x(x-1)(x-2) \frac{n!}{x(x-1)(x-2)(x-3)(x-4)\dots(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=4}^n x(x-1)(x-2)(x-3) \frac{n!}{x(x-1)(x-2)(x-3)(x-4)\dots(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=4}^n \frac{n!}{(x-4)!(n-x)!} p^4 p^{x-4} q^{n-x}$$

$$= \sum_{x=4}^n \frac{n(n-1)(n-2)(n-3)(n-4)!}{(x-4)!(n-x)!} p^4 p^{x-4} q^{n-x}$$

$$= n(n-1)(n-2)(n-3) p^4 \sum_{x=4}^n \frac{(n-4)!}{(x-4)!(n-x)!} p^{x-4} q^{n-x}$$

$$= n(n-1)(n-2)(n-3) p^4 \sum_{x=4}^n \binom{n-4}{x-4} p^{x-4} q^{n-x}$$

$$= n(n-1)(n-2)(n-3) p^4 (1)$$

$$= n(n-1)(n-2)(n-3) p^4$$

Now from eqn (A) we get

$$E(x^4) = \left[\sum_{x=0}^n x(x-1)(x-2) + \frac{1}{2} \sum_{x=0}^n x(x-1)(x-2) + \sum_{x=0}^n x(x-1) + \sum_{x=0}^n x \right] p^4 q^{n-x} \quad \text{--- (B)}$$

By putting values in eqn (2)

$$\mu'_2 = n(n-1)p^2 + 6n(n-1)p^3 + 7n(n-1)p^4 + \dots$$

Moments About Mean

1st Moment:-

As we know that,

$$\mu_1 = \mu'_1 - n\mu = 0$$

$$= np - np$$

$$= np - np$$

$$\mu_1 = 0$$

2nd Moment:-

As we know that:

$$\mu_2 = \mu'_2 - (n\mu)^2$$

$$\begin{aligned}
 \text{Ex. } \mu_2 &= \sum_{i=1}^n i^2 p + n p - n p^2 \\
 &= (n^2 - n p^2 + n p - n^2 p^2) \\
 &= n^2 p^2 - n p^2 + n p - n^2 p^2 \\
 &= n^2 p^2 - n^2 p^2 - n p^2 + n p \\
 &= n p - n p^2 \\
 &= n p (1 - p)
 \end{aligned}$$

$$\mu_2 = n p q \quad \text{As } (1 - p) = q$$

3rd Moment:-

As we know that.

$$\begin{aligned}
 \mu_3 &= \mu_3' - 3\mu_2' \mu_1' + 2(\mu_1')^3 \\
 &= n(n-1)(n-2)p^3 + 3n(n-1)p^2 + n p - \\
 &\quad 3np(n(n-1)p^2 + n p) + 2n^3 p^3 \\
 &= (n^3 - n^2 - n)(n-2)p^3 + 3n^2 p^2 - 3np^2 + n p \\
 &\quad - 3np(n^2 - n)p^2 + n p + 2n^3 p^3 \\
 &= (n^3 - 2n^2 - n^2 + 2n)p^3 + 3n^2 p^2 - 3np^2 + n p \\
 &\quad - 3n^3 p^3 + 3n^2 p^3 + n p + 2n^3 p^3
 \end{aligned}$$

$$\begin{aligned}
 &= r^3 p^3 - 2r^2 p^3 - r^2 p^2 + 2r^2 p^3 - r^2 p^2 - 2r^2 p^3 \\
 &= -7r^2 p^3 + 3r^2 p^3 - 3r^2 p^3 + 2r^2 p^3 \\
 &= -3r^2 p^3 - 7r^2 p^3 - 3r^2 p^3 + 2r^2 p^3 \\
 &= -2r^2 p^3 - 3r^2 p^3 - r^2 p^3 \\
 &= -2r^2 p^3 - 3r^2 p^3 - r^2 p^3 \\
 &= -r^2 p (2p^3 - 3p^3 - 1) \\
 &= -r^2 p (1 - 3p^3 + 2p^3) \\
 &= -r^2 p (1 - 2p^3 - p^3 + 2p^3) \\
 &= -r^2 p (1 - 2p^3) - p (1 - 2p^3) \\
 &= -r^2 p (1 - p) (1 - 2p^3) \\
 &= -r^2 p (q (1 - 2p^3)) \\
 \mu_3 &= -r^2 p q (1 - 2p^3)
 \end{aligned}$$

4th Moment:-

As we know that:

$$\begin{aligned}
 \mu_4 &= \mu_4' - 4\mu_1' \mu_3' + 6(\mu_2')^2 (\mu_1') - 3(\mu_1')^4 \\
 &= [r^4 p^4 - 6r^3 p^4 + 11r^2 p^4 - 6r p^4 + 6r^3 p^3 - 12r^2 p^3 \\
 &\quad + 12r p^3 - 7r^2 p^3 - 1r p^3 + r p^3] - [4r^3 p^3 - 12r^2 p^3 \\
 &\quad + 8r^2 p^3 + 12r^2 p^3 - 12r^2 p^3 + 4r^2 p^3] - [6r^3 p^3 - \\
 &\quad 6r^3 p^3 + 6r^3 p^3] - 3r^4 p^4
 \end{aligned}$$

$$\begin{aligned}
 L_4 &= 1p^4 - 6np^3 + 11n^2p^2 - 6n^3p + 6n^4 - 18n^3p^3 \\
 &\quad + 12n^2p^2 + 7n^3p^2 - 7np^4 + np^5 + 4n^2p^4 + 12n^3p^3 \\
 &\quad + 8n^4p^2 - 12n^5p^2 + 12n^6p^2 - 4n^7p^2 + 6n^8p^2 + n^9 \\
 &\quad + 6n^3p^3 - 3n^4p^4
 \end{aligned}$$

$$\begin{aligned}
 &+ n^5 [p^4 - 11p^3 + 6p^2 - 3p] + n^6 [-6p^3 + 6p^2 \\
 &\quad + 12p^2 - 2p] + n^7 [11p^4 + 7p^3 - 18p^2 - 8p^2 + 12p^2 - \\
 &\quad 4p^2] + n^8 [-6p^3 + 12p^2 - 7p^2 + p]
 \end{aligned}$$

$$\begin{aligned}
 &= n^4 [p^4 - 4p^3 - 3p^2] + n^5 [-6p^3 + 12p^2 - 6p^2 - 4p^2 \\
 &\quad - 12p^3 + 6p^2] + n^6 [11p^4 - 6p^3 - 18p^2 + 12p^2 - 7p^2 \\
 &\quad - 4p^2] + np [-6p^3 + 12p^2 - 7p + 1]
 \end{aligned}$$

$$\begin{aligned}
 &= n^4 [p^4 - p^4] + n^5 [6p^4 - 6p^4 - 6p^3 + 6p^3] + n^6 \\
 &\quad [3p^4 - 6p^3 + 3p^2] + np [-6p^3 + 12p^2 - 7p + 1]
 \end{aligned}$$

$$\begin{aligned}
 &= 0 + 0 + n^2 \cdot 3p^2 [p^2 - 2p + 1] + np [-6p^3 + 12p^2 \\
 &\quad - 7p + 1]
 \end{aligned}$$

$$= 3n^2p^2(1-p)^2 + np(1-p)(1-6p+6p^2)$$

$$= 3n^2p^2q^2 + npq(1-6p+6p^2)$$

$$= 3n^2p^2q^2 + npq(1-6p(1-p))$$

$$- 3n^2p^2q + npq(1 - 6pq)$$

$$- 3np^2q + npq(3npq + 6 - 6pq)$$

$$- npq(3npq + 1 - 6pq)$$

$$- npq(1 + 3npq - 6pq)$$

$$M_2 - npq(1 + 3pq + n - 2)$$

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Kurtosis of Binomial Distribution

$$\text{Kurtosis} = \frac{\mu_4}{\mu_2^2}$$

By putting the values we get.

$$= \frac{npq(1+3pq(n-2))}{(npq)^2}$$

$$= \frac{npq(1+3pq(n-2))}{(npq)^2}$$

$$\text{Kurtosis} = \frac{1+3pq(n-2)}{npq}$$

Skewness of Binomial Distribution

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

By putting the values we get.

$$= \frac{npq(1-2p)}{(npq)^{3/2}}$$

$$= \frac{npq(1-2p)}{(npq)^{1/2} \cdot (npq)^{1/2}}$$

Skema ini =

$$\frac{b \cdot d \cdot f}{1 - 2P}$$

$$= \frac{b \cdot d \cdot f \cdot (1 - 2P)}{1 - 2P}$$

و كذا في باقي العناصر

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Moment Generating Functions

Binomial Distribution

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

$$x=0, 1, 2, \dots, n$$

$$E(X) = np$$

M.G.F of Binomial Distribution

n, p parameters

$$(p+q)=1$$

$$M_0(t) = E(e^{tx})$$

$$= \sum_{x=0}^n \binom{n}{x} p^x q^{n-x} e^{tx}$$

$$M_0(t) = (q + pe^t)^n$$

1st Moment:-

$$M_0'(t) = n(q + pe^t)^{n-1} (0 + pe^t)$$

$$= npe^t (q + pe^t)^{n-1}$$

$$\text{put } t = 0$$

$$= npe^0 (q + pe^0)^{n-1}$$

$$= np(1) (q + p)^{n-1}$$

$$= np(1)^{n-1}$$

$$\because q + p = 1$$

$$\mu_1' = E(X) = np$$

2nd Moment:-

$$\begin{aligned}
 M_2(t) &= n p e^t (q + p e^t)^{n-1} \\
 &= n p [e^t (q + p e^t)^{n-1} + e^t (n-1) (q + p e^t)^{n-2} (0 + p e^t)] \\
 &= n p e^t [(q + p e^t)^{n-1} + (n-1) p e^t (q + p e^t)^{n-2}] \\
 \text{put } t &= 0 \\
 &= n p e^0 [(q + p e^0)^{n-1} + (n-1) p e^0 (q + p e^0)^{n-2}] \\
 &= n p [(q + p)^{n-1} + (n-1) p (q + p)^{n-2}] \quad \because e^0 = 1 \\
 &= n p [1^{n-1} + (n-1) p (1^{n-2})] \quad \because p + q = 1 \\
 &= n p [(n-1) p]
 \end{aligned}$$

$$M_2'(x^2) = n p + n(n-1) p^2$$

3rd Moment:-

$$\begin{aligned}
 M_3(t) &= n p e^t [(q + p e^t)^{n-1} + (n-1) p e^t (q + p e^t)^{n-2}] \\
 &\quad + n(n-1) p^2 e^t (q + p e^t)^{n-2} + n(n-1) p^2 e^t (n-2) (q + p e^t)^{n-3} (0 + p e^t) \\
 &\quad + n(n-1) p^2 e^t (n-2) p e^t (q + p e^t)^{n-3} \\
 \text{put } t &= 0 \\
 &= n p e^0 [(q + p e^0)^{n-1} + n p e^0 (n-1) (q + p e^0)^{n-2} (p e^0) + n(n-1) p^2 e^0 (q + p e^0)^{n-2} \\
 &\quad + n(n-1) p^2 e^0 (n-2) p e^0 (q + p e^0)^{n-3} (p e^0)] \\
 \because e^0 &= 1, \quad q + p = 1
 \end{aligned}$$

$$\begin{aligned}
 &= n p [1^{n-1} + n p (n-1) 1^{n-2} (p) + n(n-1) p^2 1^{n-2} (p) \\
 &\quad + n(n-1) p^2 (n-2) 1^{n-3} (p)] \\
 &= n p^3 + n(n-1) p^3 + 2n(n-1) p^3 + n(n-1)(n-2) p^3
 \end{aligned}$$

$$M_3'(x^3) = n p + 3n(n-1) p^3$$

4th Moment:-

$$\begin{aligned}
 & n p e^1 (q + p e^1)^{n-1} + n p e^1 (n-1) 2 q + p e^1)^{n-2} (p e^1) + n(n-1) \\
 & p^2 e^2 (q + p e^1)^{n-2} + n(n-1) p^2 e^2 (n-2) (q + p e^1)^{n-3} (p e^1) \\
 & = n p e^1 (q + p e^1)^{n-1} + n(n-1) p^2 e^2 (q + p e^1)^{n-2} + n(n-1) p^2 e^2 \\
 & (q + p e^1)^{n-3} + n(n-1)(n-2) p^3 e^3 (q + p e^1)^{n-3}
 \end{aligned}$$

$$\begin{aligned}
 M_4'(u) &= [n p e^1 (q + p e^1)^{n-1} + n p e^1 (n-1) 2 q + p e^1)^{n-2} (p e^1) + \\
 & [n(n-1) p^2 e^2 (q + p e^1)^{n-2} + n(n-1) p^2 e^2 (n-2) (q + p e^1)^{n-3} \\
 & (p e^1)] + [n(n-1) p^2 e^2 + (q + p e^1)^{n-1} + n(n-1) p^2 e^2 \\
 & (n-1) (q + p e^1)^{n-2} (p e^1)] + [n(n-1)(n-2) p^3 e^3 \\
 & (q + p e^1)^{n-2} + n(n-1)(n-2) p^3 e^3 (n-3) (q + p e^1)^{n-4} (p e^1)] \\
 & \text{put } u = 1
 \end{aligned}$$

$$\begin{aligned}
 & = n p (1^{n-1} + n p (n-1) (1)^{n-2} (p) + n(n-1) p^2 (2) (1)^{n-2} + n(n-1) \\
 & p^2 (n-2) (1)^{n-3} (p) + n(n-1) 4 p^2 (1)^{n-1} + n(n-1) 2 p^2 (n-2) \\
 & (1)^{n-3} (p) + n(n-1)(n-2) p^3 (3) (1)^{n-3} + n(n-1)(n-2) p^3 \\
 & (n-3) (1)^{n-4} (p)
 \end{aligned}$$

$$\begin{aligned}
 & = n p + n(n-1) p^2 + 2 n(n-1) p^2 + n(n-1)(n-2) p^3 + 4 n(n-1) p^3 \\
 & + n(n-1)(n-2) p^3 + 3 n(n-1)(n-2) p^3 + n(n-1)(n-2)(n-3) p^4 \\
 & = n p + n(n-1) p^2 + 2 n(n-1) p^2 + 4 n(n-1) p^3 + n(n-1)(n-2) p^3 + \\
 & 2 n(n-1)(n-2) p^3 + 3 n(n-1)(n-2) p^3 + n(n-1)(n-2)(n-3) p^4
 \end{aligned}$$

$$U.F(Y^4) = n p + 7 n(n-1) p^2 + 6 n(n-1)(n-2) p^3 + n(n-1)(n-2)(n-3) p^4$$

Moments About Mean

1st Moment:-

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$np - np = 0$$

$$np - np = 0$$

2nd Moment:-

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$= np + n(n-1)p^2 - (np)^2$$

$$= np + n^2p^2 - np^2 - n^2p^2$$

$$= np[1 + np - p - np]$$

$$= np[1 - p + np - np]$$

$$= np(1-p)$$

$$\because 1-p = q$$

$$= npq$$

3rd Moment:-

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$= np + 3n(n-1)p^2 + n(n-1)(n-2)p^3 - 3(np + n(n-1)p^2)$$
$$+ 2(np)^3$$

$$\begin{aligned}
 \mu_2 &= np + (3n^2 - 3n)p^2 - 2(n-1)(np^2 - 2p^3) - 3(np + n^2p^2 + np^3)p \\
 &= np + 3n^2p^2 - 3np^2 + n^3p^3 - 2n^2p^3 - n^2p^3 + 2np^3 - 3(n^2p^2 + n^3p^3 - n^2p^3) + 2n^3p^3 \\
 &= np + 3n^2p^2 - 3np^2 + n^3p^3 - 2n^2p^3 - n^2p^3 + 2np^3 - 3n^2p^2 + 3n^2p^3 + 2n^3p^3 \\
 &= np - 3np^2 + 2np^3 \\
 &= np(1 - 3p + 2p^2) \\
 &= np(1-p)(1-2p) \\
 &= npq(1-p) \\
 \mu_3 &= npq(1-p)
 \end{aligned}$$

4th Moment:-

$$\begin{aligned}
 \mu_4 &= \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4 \\
 &= n(n-1)(n-2)(n-3)p^4 + 6n(n-1)(n-2)p^3 + 7n(n-1)p^2 + np - 4(np)(n(n-1)(n^2-2p^2)) + 3n(n-1)p^2 + np + 6(np)^2(n(n-1)p^2 + np) - 3(np)^4 \\
 &= n^4(p^4 - q^4) + n^3(-6p^4 + 6p^3 + 6p^3 - 6p) + n^2(11p^4 - 18p^3 + 7p^2 - 4p^2 + 18p^3 - 8p^4) + n(-6p^4 + 12p^3 - 7p^2 + p) \\
 &= 3n^3p^3(1-p)^2 + np(1-p)(1-6p+6p^2) \\
 &= 3n^3p^3(q)^2 + np(q)(1-6p+6p^2) \\
 &= 3n^3p^3q^2 + npq(1-6p+6p^2) \\
 &= npq[3npq + 1 - 6pq] \\
 &= npq[1 + 3(n-2)pq] \\
 &= npq[1 + 3(n-2)pq]
 \end{aligned}$$

Skewness

$$\text{Skewness} = \frac{(u_3)^2}{(u_2)^3} = \beta_1$$

By putting values we get

$$= \frac{[npq(q-p)]^2}{(npq)^3}$$

$$= \frac{n^2 p^2 q^2 (q-p)^2}{n^3 p^2 q^2}$$

$$\text{Skewness} = \frac{(q-p)^2}{npq} = \beta_1 = \frac{(1-2p)^2}{npq} = \frac{1-2p}{\sqrt{npq}}$$

Kurtosis

$$\text{Kurtosis} = \frac{u_4}{(u_2)^2} = \beta_2$$

By putting values we get

$$= \frac{npq[1+3pq(n-2)]}{(npq)^2}$$

$$\beta_2 = \frac{npq[1+3pq(n-2)]}{n^2 p^2 q^2} = \frac{1+3pq(n-2)}{npq}$$

Probability Generating Functions

$$G = E(\theta^x) = \sum_{x=0}^n \theta^x \cdot p(x)$$

For Binomial Distribution.

$$P(X=x) = \binom{n}{x} p^x \cdot q^{n-x}$$

$$= \binom{n}{0} p^0 q^n + \binom{n}{1} p^1 q^{n-1} + \binom{n}{2}$$

$$p^2 q^{n-2} + \binom{n}{1} p^3 q^{n-3} + \dots$$

$$+ \binom{n}{n} p^n q^0$$

$$= (q+p)^n$$

$$G = E(\theta^x) = \sum_{x=0}^n \theta^x \cdot \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^n \binom{n}{x} (p\theta)^x q^{n-x}$$

$$G = (q + p\theta)^n$$

Prob. generating function of B.D.

$$G' = \frac{\partial G}{\partial \theta} = n(q+p\theta)^{n-1}(0+p)$$

$$= np(q+p\theta)^{n-1}$$

$$\text{put } \theta = 1$$

$$m'_1 = np(q+p)^{n-1}$$

$$m'_1 = np$$

$$G'' = np(n-1)(q+p\theta)^{n-2}(0+p)$$

$$= n(n-1)p^2(q+p\theta)^{n-2}$$

$$\text{put } \theta = 1$$

$$G'' = n(n-1)p^2(1)^{n-2}$$

$$m''_2 = n(n-1)p^2$$

$$G''' = n(n-1)p^2(q+p\theta)^{n-3}(0+p)$$

$$= n(n-1)(n-2)p^3(q+p\theta)^{n-3}$$

$$\text{put } \theta = 1$$

$$= n(n-1)(n-2)p^3$$

$$G^{iv} = n(n-1)(n-2)p^3(n-3)(q+p\theta)^{n-4}(0+p)$$

$$= n(n-1)(n-2)(n-3)p^4(q+p\theta)^{n-4}$$

$$\text{put } \theta = 1$$

$$= n(n-1)(n-2)(n-3)p^4$$

Factorial Generating Function

Binomial Distribution

$$M.G.F = M_0(t) = (q + pe^t)^n$$

put $e^t = 1 + \alpha$ we get.

$$H_0 = [q + p(1 + \alpha)]^n$$
$$= (q + p + p\alpha)^n$$

1st:

$$H_0' = n(q + p + p\alpha)^{n-1} (p)$$

$$= np(q + p + p\alpha)^{n-1}$$

put $\alpha = 0$

$$= np(1 + 0)^{n-1}$$

$$\because (q + p) = 1$$

$$= np(1)^{n-1}$$

$$E(x) = np$$

$$np(q + p + p\alpha)^{n-1}$$

$$H_0'' = np(n-1)(q + p + p\alpha)^{n-2} (p)$$

$$= n(n-1)p^2(q + p + p\alpha)^{n-2}$$

put $\alpha = 0$

$$= n(n-1)p^2(1 + 0)^{n-2}$$

$$= n(n-1)p^2(1)^{n-2}$$

$$= n(n-1)p^2$$

$$n(n-1)p^2(q+p+p\alpha)^{n-2}$$

$$H_0'' = n(n-1)p^2(n-2)(q+p+p\alpha)^{n-3}(p)$$

$$= n(n-1)(n-2)p^3(q+p+p\alpha)^{n-3}$$

$$\text{put } \alpha = 0$$

$$= n(n-1)(n-2)p^3(q+p)^{n-3}$$

$$= n(n-1)(n-2)p^3(1)^{n-3}$$

$$= n(n-1)(n-2)p^3$$

$$E[x(x-1)(x-2)] = n(n-1)(n-2)p^3$$

$$n(n-1)(n-2)p^3(q+p+p\alpha)^{n-3}$$

$$H_0^{iv} = n(n-1)(n-2)p^3(n-3)(q+p+p\alpha)^{n-4}(p)$$

$$= n(n-1)(n-2)(n-3)p^4(q+p+p\alpha)^{n-4}$$

$$\text{put } \alpha = 0$$

$$= n(n-1)(n-2)(n-3)p^4(q+p)^{n-4}$$

$$= n(n-1)(n-2)(n-3)p^4(1)^{n-4}$$

$$= n(n-1)(n-2)(n-3)p^4$$

$$E[x(x-1)(x-2)(x-3)] = n(n-1)(n-2)(n-3)p^4$$

Now we find ordinary moments

Moments About Origin

1st # $\mu'_1 = E(x) = np$

2nd # $\mu'_2 = E(x^2) = E[x(x-1)] + E(x)$
 $= n(n-1)p^2 + np$

3rd # $\mu'_3 = E(x^3) = E[x(x-1)(x-2)] + 3E[x(x-1)] + E(x)$
 $= n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np$

4th # $\mu'_4 = E(x^4) = E[x(x-1)(x-2)(x-3)] + 6E[x(x-1)(x-2)] + 7E[x(x-1)] + E(x)$
 $= n(n-1)(n-2)(n-3)p^4 + 6n(n-1)(n-2)p^3 + 7n(n-1)p^2 + np$

Moments About Mean

1st # $\mu_1 = \mu'_1 - \mu'_1 = 0$

2nd # $\mu_2 = \mu'_2 - (\mu'_1)^2$

$$= n(n-1)p^2 + np - n^2p^2$$

$$= (n^2 - n)p^2 + np - n^2p^2$$

$$= \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2}$$

$$= np \cdot np^2$$

$$= np(1-p)$$

$$\mu_2 = npq$$

$$\because (1-p) = q$$

$$\begin{aligned} \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= [n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np] - 3[(n(n-1) + np)(np)] + 2(np)^3 \\ &= (n^3 - n^2 - n^2 + 2n)p^3 + (3n^2 - 3n)p^2 + np - 3[(n^2 - n)p^2 + np^2] + 2n^3p^3 \\ &= (n^3 - 2n^2 - n^2 + 2n)p^3 + 3n^2p^2 - 3np^2 + np - 3(n^2p^2 - np^2 + np^2) + 2n^3p^3 \\ &= n^3p^3 - 2n^2p^3 - n^2p^3 + 2np^3 + 3n^2p^2 - 3np^2 + np - 3n^2p^2 + 3np^2 - 3np^2 + np + 2n^3p^3 \\ &= [n^3p^3 - n^2p^3 + n^2p^2] + 2n^3p^3 \\ &= n^3p^3 - 2n^2p^3 - n^2p^3 + 2np^3 + 3n^2p^2 - 3np^2 + np \\ &= -3n^3p^3 + 3n^2p^3 - 3n^2p^2 + 2n^3p^3 \\ &= -\cancel{1}n^3p^3 + 2np^3 - 3np^2 + np + 2n^3p^3 \\ &= 2np^3 - 3np^2 + np \\ &= np(2p^2 - 3p + 1) \end{aligned}$$

$$\begin{aligned}
 &= np(1-3p+2p^2) \\
 &= np(1-p-2p+2p^2) \\
 &= np(1-2p-p+2p^2) \\
 &= np[1(1-2p)-p(1-2p)] \\
 &= np(1-2p)(1-p) \\
 &= np(1-2p)(q) \quad \because 1-p=q
 \end{aligned}$$

$$\mu_3 = npq(1-2p)$$

4th :

$$\begin{aligned}
 \mu_4 &= \mu_4' - 4\mu_3'\mu_1' + 6(\mu_2')^2(\mu_1') - 3(\mu_1')^4 \\
 &= [n(n-1)(n-2)(n-3)p^4 + 6n(n-1)(n-2)p^3 + 7n(n-1)p^2 + np] - 4[(n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np)(np)] + 6 \\
 &\quad (np)^2(n(n-1)p^2 + np) - 3(np)^4 \\
 &= [n^4p^4 - 6n^3p^4 + 11n^2p^4 - 6np^4 + 6n^3p^3 - 18n^2p^3 + 12np^3 + 7n^2p^2 - 7np^2 + np] - [4n^4p^4 - 12n^3p^4 + 8n^2p^4 + 12n^3p^3 - 12n^2p^3 + 4np^3] + [6n^4p^4 - 6n^3p^4 + 6n^3p^3] - 3n^4p^4 \\
 &= n^4[p^4 - 4p^4 + 6p^4 - 3p^4] + n^3[-6p^4 + 6p^3 + 12p^4 - 12p^3] + n^2[11p^4 - 8p^4 - 18p^3 + 12p^3 + 7p^2 - 4p^2] + np[-6p^3 + 12p^2 - 7p + 1] \\
 &= n^4[p^4 - p^4] + n^3[6p^4 - 6p^4 - 6p^3 + 6p^3] + n^2[3p^4 - 6p^3 + 3p^2] + np[-6p^3 + 12p^2 - 7p + 1]
 \end{aligned}$$

$$= 0 + 0 + n^2 \cdot 3p^2 [p^2 - 2p - 1] + np [6p^3 + 12p^2 - 7p + 1]$$

$$= 3n^2 p^2 [1 - p]^2 + np (1 - p) (1 - 6p + 6p^2)$$

$$= 3n^2 p^2 q^2 + npq (1 - 6p + 6p^2)$$

$$= 3n^2 p^2 q^2 + npq (1 - 6p + 6p^2)$$

$$= 3n^2 p^2 q^2 + npq (1 - 6pq)$$

$$= npq [3npq + (1 - 6pq)]$$

$$= npq [3npq + 1 - 6pq]$$

$$= npq [1 + 3npq - 6pq]$$

$$K = npq (1 + 3pq (n - 2))$$

Factorial Cumulative Generating Functions

Binomial Distribution

F.G.F. of Binomial Distribution

Here $(1+px)^n$

By taking log of Factorial G.F.

$$\begin{aligned}h_0(x) &= \log[(1+px)^n] \\&= n \log(1+px)\end{aligned}$$

By taking derivative w.r.t x we get

1st $\rightarrow$

$$h'_0(x) = n \cdot \frac{1}{1+px} \cdot p$$

$$= np \cdot \frac{1}{1+px}$$

$$p(x) \quad x = 0$$

$$h_1(\omega) = rp \frac{1}{1+p\omega}$$

$$h_0(\omega) = rp = k_0(\omega)$$

2nd:- $rp \frac{1}{1+p\omega} = rp(1+p\omega)^{-1}$

$$h_0'(\omega) = rp \frac{d}{d\omega} (1+p\omega)^{-1} (p^1)$$

$$= -rp^2 (1+p\omega)^{-2}$$

put $\omega = 0$

$$= -rp^2 (1+p(0))^{-2}$$

$$k_{res} = -rp^2$$

3rd:- $-rp^2 (1+p\omega)^{-2}$

$$h_0'(\omega) = -rp^2 (-2) (1+p\omega)^{-3} \cdot p$$

$$= 2rp^3 (1+p\omega)^{-3}$$

put $\omega = 0$

$$= 2rp^3 (1+p(0))^{-3}$$

$$k_{res} = 2rp^3$$

4th =

$$2np^3(1+p\alpha)^{-3}$$

$$k_2(\alpha) = 2np^3(-3)(1+p\alpha)^{-4} \cdot p$$

$$= -6np^4(1+p\alpha)^{-4}$$

$$\text{put } \alpha = 0$$

$$= -6np^4(1+p(0))^{-4}$$

$$= -6np^4(1)^{-4}$$

$$k_2(0) = -6np^4$$

Cumulents (Kapas)

1st

$$K_1 = k_1(n) = m_1'$$

$$= np = m_1'$$

2nd

$$K_2 = k_2(n) + k_2(2) = \mu_2$$

$$= np - np^2$$

$$= np(1-p)$$

$$= npq = \mu_2$$

$$\because (1-p) = q$$

3rd

$$K_3 = k_{100} + 3k_{101} + k_{111} = 11$$

$$= np + 3(-np^2) + np^3$$

$$= np + 3np^2 + 2np^3$$

$$= np(1 + 3p + 2p^2)$$

$$= np(1 + 2p + p + 2p^2)$$

$$= np(1 + 2p) + p(1 + 2p)$$

$$= np(1 + p)(1 + 2p)$$

$$= npq(1 + 2p) = 11 \quad \because (1 + p) = q$$

4th

$$K_4 = k_{1000} + 7k_{1001} + 6k_{1011} + k_{1111}$$

$$= np + 7(-np^2) + 6(2np^3) + (-6np^4)$$

$$= np + 7np^2 + 12np^3 - 6np^4$$

$$= np[1 + 7p + 12p^2 - 6p^3]$$

Now

$$11_q = K_4 + 3K_3$$

$$= np[1 + 7p + 12p^2 - 6p^3] + 3np^2q^2$$

$$= np[1 + p(7 + 12p + 6p^2)] + 3np^2q^2$$

$$\begin{aligned}
&= np[1-p(7-12p+6p^2)] + 3n^2p^2q^2 \\
&= np[1-p(1+6-12p+6p^2)] + 3n^2p^2q^2 \\
&= np[1-p(1+(6-6p-6p+6p^2))] + 3n^2p^2q^2 \\
&= np[1-p(1+(6(1-p)-6(1-p)))] + 3n^2p^2q^2 \\
&= np[1-p(1+(1-p)(6-6p))] + 3n^2p^2q^2 \\
&= np[1-p(1+(1-p)6(1-p))] + 3n^2p^2q^2 \\
&= np[1-p(1+q \cdot 6q)] + 3n^2p^2q^2 \\
&= np[1-p(1+6q^2)] + 3n^2p^2q^2 \\
&= np[1-p-6pq^2] + 3n^2p^2q^2 \\
&= np[q-6pq^2] + 3n^2p^2q^2 \\
&= npq + 6np^2q^2 + 3n^2p^2q^2 \\
&= npq[1+6pq+3npq] \\
&= npq[1+3npq-6pq] \\
&= npq[1+3pq(n-2)] = \mu_4
\end{aligned}$$

Skewness

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

By putting the values.

$$Skewness = \frac{\sum f_p q (1-2p)}{(\sum f_p q)^{3/2}}$$

$$= \frac{\sum f_p q^3}{(\sum f_p q)^{3/2}}$$

$$\frac{\sum f_p q (1-2p)}{(\sum f_p q)^{3/2}}$$

$$= \frac{\sum f_p q^3}{(\sum f_p q)^{3/2}}$$

$$= \frac{\sum f_p q (1-2p)}{(\sum f_p q)^{3/2}}$$

$$\frac{\sum f_p q (1-2p)}{(\sum f_p q)^{3/2}}$$

$$\frac{1-2p}{\sum f_p q}$$

Kurtosis

$$Kurtosis = \frac{\mu_4}{\mu_2^2}$$

$$\mu_4 = \sum f_p q^4$$

$$= \frac{\sum f_p q (1+3pq(n-2))}{(\sum f_p q)^2}$$

$$= \frac{\sum f_p q (1+3pq(n-2))}{(\sum f_p q)^2}$$

$$= \frac{1+3pq(n-2)}{\sum f_p q}$$

Cumulant Generating Function

Binomial Distribution:

Moment Generating Function of
Binomial Distribution:

$$M_0(t) = (q + pe^t)^n$$

By taking log of M.G.F. we get

$$\log M_0(t) = n \log(q + pe^t)$$

By taking derivative w.r.t t we get.

1st:

$$\begin{aligned} C'_0(t) &= n(q + pe^t)^{-1} \cdot pe^t \\ &= npe^t(q + pe^t)^{-1} \end{aligned}$$

$$\text{put } t=0$$

$$= npe^0(q + pe^0)^{-1}$$

$$= np(q + p)^{-1}$$

$$= np(1)^{-1}$$

$$\because e^0 = 1$$

$$\because q + p = 1$$

$$K_1 = np$$

2nd order differential equation

$$np e^t (q + p e^t)^{-1}$$

$$C_0''(t) = np [e^t (q + p e^t)^{-1} + e^t (-1) (q + p e^t)^{-2} \cdot p e^t]$$

$$= np [e^t (q + p e^t)^{-1} - p e^{2t} (q + p e^t)^{-2}]$$

$$= np e^t (q + p e^t)^{-2} [(q + p e^t) - p e^t]$$

$$= np e^t (q + p e^t)^{-2} [q + p e^t - p e^t]$$

$$= np q e^t (q + p e^t)^{-2}$$

$$p w \quad t = 0$$

$$= np q e^0 (q + p e^0)^{-2}$$

$$= np q (q + p)^{-2}$$

$$K_1 = np q$$

3rd:-

$$np q e^t (q + p e^t)^{-2}$$

$$C_0'''(t) = np q [e^t (q + p e^t)^{-2} + e^t (-2) (q + p e^t)^{-3} \cdot p e^t]$$

$$= np q [e^t (q + p e^t)^{-2} - 2 p e^{2t} (q + p e^t)^{-3}]$$

$$= np q e^t (q + p e^t)^{-3} [(q + p e^t) - 2 p e^t]$$

$$= np q e^t (q + p e^t)^{-3} [q + p e^t - 2 p e^t] (-p e^t)$$

$$= np q e^t (q + p e^t)^{-3} [q - p e^t] (-p e^t)$$

$$= npq e^t (q + pe^t)^{-3} (q - pe^t)$$

$$\text{put } t = 0$$

$$= npq e^0 (q + pe^0)^{-3} (q - pe^0)$$

$$= npq (q + p)^{-3} (q - p)$$

$$= npq (1)^{-3} (q - p)$$

$$\because q + p = 1$$

$$K_3 \therefore = npq (q - p)$$

4th - $npq^2 e^t (q + pe^t)^{-3} - npq^2 e^{2t} (q + pe^t)^{-3} (pe^t)$

$$C_0''(t) = [npq^2 e^t (q + pe^t)^{-3} + npq^2 e^t (-3)(q + pe^t)^{-4}]$$

$$- [np^2 q (2) e^{2t} (q + pe^t)^{-3} + np^2 q e^{2t} (-3)(q + pe^t)^{-4} (pe^t)]$$

$$= npq^2 e^t (q + pe^t)^{-3} - 3npq^2 e^{1t} (q + pe^t)^{-4}$$

$$- 2np^2 q e^{2t} (q + pe^t)^{-3} + 3np^3 q e^{3t} (q + pe^t)^{-4}$$

$$\text{put } t = 0$$

$$= npq^2 (q + p)^{-3} - 3npq^2 e^0 (q + p)^{-4} - 2np^2 q$$

$$(q + p)^{-3} + 3np^3 q (q + p)^{-4}$$

$$= npq^2 - 3npq^2 - 2np^2 q + 3np^3 q$$

$$= npq (q^2 - 3pq - 2p + 3p^2)$$

$$K_4 = npq [q(1 - 3p) + p(-2 + 3p)]$$

$$m_y = K_y + 3K_z^2$$

$$= npq [q(1-3p) + p(-2+3p)] + 3n^2 p^2 q^2$$

$$= npq [q(1-3p) + p(-2+3p) + 3npq]$$

$$= npq [q-3pq - 2p + 3p^2 + 3npq]$$

$$= npq [q-2p+3p^2-3pq(1-n)]$$

$$= npq [q-p(2-3p)-3pq(1-n)]$$